

When to Escalate: A Cost-Aware Belief Policy for Conversational Agents Under Hidden Intent

Kapardhi Kannekanti
Independent Researcher
Hyderabad, India
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Abstract

A conversational agent replying to inbound sales leads must choose an action before it knows the lead’s intent. It can answer, ask a qualifying question, hold, or escalate to a human, but the true buying-readiness and whether a human is actually needed are hidden. The two ways of being wrong are not equally costly. A missed escalation, where a lead who needed a person did not get one, is far more expensive than a needless one, and some wrong answers are not a cost to trade off at all but a line the agent must not cross. We build a small agent that makes this trade-off explicit. It holds a belief over the hidden state, split into readiness (hot, warm, cold) and a separate probability that the case needs a human, and at each message it picks the action with the lowest expected cost under a cost matrix that encodes the asymmetry. Escalation is split into notify, where a human is told while the conversation continues, and pause, where the agent stops, because the two carry different costs. The hardest wrong answers, such as false legal or land claims, are enforced as a hard constraint rather than a priced term.

We test on 100 synthetic cases with cached language-model beliefs, comparing the cost-aware policy against the same policy with a uniform cost matrix and against three fixed-action baselines. Two of those baselines give opposite answers, so both are named wherever the cost result appears. Against the uniform-cost version of the same rule — same belief, same feasible set, only the asymmetry removed — the cost-aware policy reaches 1.72 against 2.58 over all 100 cases, a reduction of a third; that gap is what cost-awareness buys, and it is against that baseline and no other. Against always-notify, which escalates every message, there is no cost win: 1.72 against 1.74 over the same 100 cases is a near-tie, a total of 172 against 174, so the cost-aware policy is not the cheaper policy and is not claimed to be. What it buys there is human load rather than cost — over all 100 cases it reaches the same cost while escalating 43 times rather than 100,

at an escalation precision of 0.605 against 0.420. The uniform-baseline gap is also not all machinery: that baseline selects the same action as a plain 0.5 threshold on all 100 cases, and reweighting the case set toward the design’s own readiness prior moves the gap from 0.86 to 1.07 rather than closing it, so the direction is robust to reweighting even as the magnitude shifts. Every missed escalation traces to one cause: the belief under-estimated the needs-human probability rather than misreading readiness. Expected calibration error on that marginal is 0.142 (bootstrap 95% CI [0.100, 0.249]), concentrated in the bins adjacent to the escalation threshold rather than the bin containing it; recalibrating that marginal on the same 100 cases is an in-sample ceiling, and the body reports it as one. Held out, the result is different in kind. Re-eliciting b_h from the model’s digit logprobs, fitting a map on 50 development cases and scoring the other 50, improves all three calibration measures named in advance — on those 50 test cases expected calibration error falls from 0.1526 to 0.0696, cross-entropy from 0.8546 to 0.8136 bits, and Brier from 0.2063 to 0.1962. Better calibration does not buy a better decision. The fitted map is isotonic, and the lowest block it pools sets a reachable-score floor of $6/23 \approx 0.2609$, above the $3/13 \approx 0.2308$ a belief must fall below for answering to be cheapest, so the threshold sits inside an interval the map cannot emit: on those same 50 cases the policy stops choosing *answer* at all, escalation precision falls from 0.667 to 0.463, and recall rises from 0.667 to 0.905. The cause is the map’s range rather than the scores it produced, and a calibration-quality gain that the range prevents from becoming a decision gain is the finding. We also report a failure the test set cannot produce by construction, a high-cost message dropped when it arrived batched with a routine one, observed in a live run. Code and data are available.

Problem statement. The agent observes an inbound message from a sales lead. It must select one action from {answer, ask a qualifying question, hold, escalate to a human} because the

lead’s true intent and buying-readiness are not known.

Note on the action set. The statement above is the original problem seed, reproduced verbatim. In the course of this work escalation was split into two distinct actions: *escalate_notify*, where a human is alerted while the conversation continues, and *escalate_pause*, where the agent stops and hands over. The two are separated because they carry materially different costs, and collapsing them would hide that difference from the decision rule. The resulting five-action set — {answer, ask, hold, *escalate_notify*, *escalate_pause*} — is the one used throughout this paper and in every reported result.

1 Introduction

A conversational agent that answers inbound sales leads has to act before it knows who it is talking to. The same short message, “can I get more info”, might come from a serious buyer, a competitor fishing for pricing, or an automated blast. The agent sees the text. It does not see the intent behind it. It still has to do something.

The two ways of getting this wrong are not equally expensive, and that is the whole problem. An agent that escalates every uncertain message to a human never misses a real buyer, but it spends human attention on messages that never needed it. An agent that answers everything itself keeps humans free, but eventually gives a confident wrong reply to a lead who needed a person, or worse, answers a question it had no business answering, like a request for legal or land documents. A naive confidence threshold treats these two failures as if they sat on the same scale. They do not. A missed hand-off to a human is far more costly than a needless one, and some wrong answers are not a cost to trade off at all but a line that must not be crossed.

The shape of that trade-off is easier to see on a decision with no agent in it. Suppose carrying an umbrella costs 2, getting caught in the rain costs 5, the forecast says 35%, and on one particular morning getting caught costs 400 instead of 5. All four of those numbers are assumed for the illustration and none is measured. Leaving the umbrella behind is cheaper while $5p < 2(1 - p)$, where p is the chance of rain, so the break-even is $p^* = 2/7 \approx 28.6\%$. At a forecast of 35% you carry, but only just, and a small error in the forecast would flip the decision. Now hold the forecast and the cost of carrying fixed, and change only what getting caught costs. On the morning of a wedding the break-even moves to $2/402 \approx 0.5\%$. The forecast has not moved and the decision is no longer close, because nothing a forecast could plausibly say would bring it back.

The general form is that the break-even is the cheap error’s cost over the sum of both errors’ costs, and it is the same expression Section 4.4 solves for this paper’s threshold, on a cost matrix rather than on an umbrella. Carrying is the false positive, a small cost paid for nothing. Getting caught is the false negative, the expensive thing that was not prevented. Escalating a lead who did not need it is the umbrella, and answering one who needed a person is the rain. What the wedding shows is what the cost matrix shows: the operating point is set by the ratio of the two errors and not by the probability,

and raising the cost of the expensive error on its own can take a decision that was marginal and make it not a question.

This paper builds a small agent that makes the trade-off explicit. It maintains a belief over the lead’s hidden state, split into readiness (hot, warm, cold) and a separate probability that the case needs a human, and at each message it chooses the action with the lowest expected cost under a cost matrix that encodes the asymmetry directly. The action set has five members: answer, ask a qualifying question, hold, *escalate_notify*, and *escalate_pause*. Escalation is split into notify and pause because the two carry different costs. A needless pause can let a live lead go cold, while a needless notify only spends a human’s glance. The hardest wrong answers, false legal or land claims, are enforced not as a high price but as a hard constraint that removes the answer action entirely.

The contribution is not a new algorithm. The decision rule is a textbook one-step Bayes decision. What this paper adds is an account of what cost-awareness actually buys on this problem and where it does not help, and one result that does not depend on the data at all: a proof that under these costs a question can never pay, together with the condition on a cost matrix under which one could. Over 100 synthetic cases with cached language-model beliefs, the cost-aware policy reaches a mean cost of 1.72 against 2.58 for the uniform-cost baseline — the same rule with every non-zero cost replaced by one, so that the belief, the feasible set and the expected-cost machinery are held fixed and only the asymmetry is removed. That is a reduction of a third, it is measured over all 100 cases, and it is against that baseline and no other. Not all of the gap is the asymmetry either. The uniform-cost baseline turns out to select the same action as a plain 0.5 threshold on the needs-human probability on all 100 cases, so part of the distance is 0.5 against the threshold the costs imply. Against always-notify, which simply escalates every message, there is no cost win at all: 1.72 against 1.74 over the same 100 cases is a near-tie, a total of 172 against 174, and what the cost-aware policy buys there is not lower cost but far less human load, reaching the same cost while escalating 43 times rather than 100 and holding an escalation precision of 0.605 against 0.420 on that same set of 100. A third comparison is less favourable still and is reported with the other two: no threshold on that probability reproduces the policy’s operating point at all, and tuning one on the 50 development cases by realised cost selects *escalate-everything*.

The failure analysis is where the design earns or loses trust. Every missed escalation traces to a single cause: the belief under-estimated the needs-human probability, and did not misread readiness. The model is measurably under-confident on that one marginal, with an expected calibration error of 0.142 on the needs-human probabilities of those 100 cases — but not in the bin the threshold falls in, which is the one bin that reads slightly over-confident. The under-confidence is in its neighbours, and the mechanism is that a large mass of cases sits pinned just below the line. So the first fix to try is recalibration rather than a redesign of the state. Section 6.9 tries it on held-out cases and reports what it buys: all three calibration measures improve, and the policy that follows stops choosing *answer* at all, because the fitted map cannot emit a probability low enough for answering to be cheapest. We

also report a failure the test set cannot even exhibit by construction, a high-cost message dropped when batched with a routine one at a turn boundary, observed in a live run and named as a limitation rather than hidden.

Two of the five actions are never chosen, and the reason is not that the case set never called for them. Scoring *ask* by the value of the information it buys yields a bound rather than a measurement: on the unconstrained action menu the value of one more question is at most $-2/13$ at every belief in the simplex. There is no narrow middle band in which asking wins, and the coarseness of the elicited belief is not what hides one — a re-elicited belief carrying 100 distinct values leaves the region exactly as empty. The bound also holds at every node of a lookahead tree, so it is not an artefact of the one-step horizon. Asking is not underpriced here; it is priced, and it loses at every depth. What transfers off this matrix is the condition behind the bound. Writing c_F and c_T for the price of a question when no human is needed and when one is, ν for a needless escalation and α for a false answer, a question can be rational at some belief only if $c_F/\nu + c_T/\alpha < 1$; these costs give $16/15$ and miss it by 6.25%. The one place the bound turns positive is the constrained menu, where the hard constraint has removed *answer*: there it is positive for $b_h < 1/5$ on the hot ray, and Section 6.3 reports that no case in this set falls inside it.

The rest of the paper is organized as follows. Section 2 places the work against learning-to-defer and selective-prediction literature and records the design changes that came from public discussion with practitioners. Section 3 gives the agent design. Section 4 gives the probability model and decision rule. Section 5 describes the test. Section 6 reports results, including Section 6.3, which proves that two of the five actions cannot be chosen under this cost matrix, and Section 6.9, which re-elicits the belief and fits a calibration map on one half of the cases to score it on the other. Section 7 gives the failure analysis and Section 8 the limitations, ethics, and human-control considerations.

2 Related work and user discussions

2.1 Related work

The escalate action places this work in the *learning-to-defer* literature, where an agent chooses between deciding itself and handing off to a human who has their own error and cost. Mozannar and Sontag [2020] give a formal, consistent treatment of this, and Madras *et al.* [2018] frame deferral with an eye to fairness and to the point that a human is not automatically the safer choice. This paper uses that framing but does not train a joint defer-classifier. The escalate action is priced in a cost matrix rather than learned, and the human is treated as a costed fallback rather than modelled with a per-case error rate.

The hold and ask actions are reject options in the sense of selective prediction, where a model acts on part of its input and abstains on the rest, trading coverage against risk [Geifman and El-Yaniv, 2017]. The policy here reports how often it hands off rather than forcing a decision, which is the coverage side of that trade.

The belief comes from a language model, so whether its probabilities can be thresholded at all depends on calibration. Guo *et al.* [2017] show that modern networks trained with a proper scoring rule are still often overconfident, and give the expected calibration error and reliability diagram used in Section 6. Their diagnosis matters here, though the sign this paper measures is the other one. What suppresses escalation is a needs-human probability read too low, which makes the automated actions look cheaper than they are; against observed frequency the belief reads under-confident on that marginal, and the calibration reported in Section 6 is of that marginal rather than of the belief as a whole.

Finally, the problem is framed as a POMDP, where the true state is hidden and the agent acts on a belief [Kaelbling *et al.*, 1998]. Exact planning over belief space is intractable, so this work solves the one-step problem myopically: it minimises immediate expected cost under the current belief and does not plan over future beliefs. This is a deliberate scope choice rather than an oversight, and what it costs depends on the cost matrix. A one-step rule carries no machinery that prices the better belief a question buys on the next turn, but on the costs used here that machinery would change no decision: Section 6.3 shows the value of a question is negative at every belief on the unconstrained action menu, and that the bound holds at every depth. Section 8 states what that does and does not license, and on what kind of matrix the horizon would cost something.

2.2 User discussions

Several design choices came from posting the problem to practitioners rather than from the literature. The record of these exchanges is in the project’s discussion log; two of them changed the design in ways that the results later bore out.

A practitioner building message agents pointed out that treating an empty text field as the trigger for escalation is fragile, because reactions, stickers, captionless media, and system events all arrive as “a message with no text” and hit the same branch. The suggested fix was to classify the event type first and choose the action from the type, rather than inferring it from the absence of text. This maps directly onto a failure the paper reports in Section 7: a callback request that arrived batched with a routine message was dropped because the two were handled as one turn. The design response, a per-message scan for critical triggers before the turn is acted on, is the same idea in a different guise: distinct signals should not be collapsed into one decision.

A second exchange, on calibration, sharpened how the belief should be checked. The point raised was that training on a proper scoring rule pushes toward calibration but does not guarantee it, so calibration has to be measured rather than assumed, and measured where the decision actually turns rather than uniformly. This is what Section 6 does: the reported calibration error is on the needs-human marginal, because that is the marginal the escalation threshold acts on. Measuring it there is also what showed the error is not where it was expected to be — the bin containing the threshold is not the under-confident one, and the under-confidence sits in its neighbours. Section 6.9 carries the same point further, fitting a calibration map on one half of the cases and scoring it on

the other.

Other exchanges shaped earlier design decisions without tying as directly to a reported result: separating “cannot parse the input” from “understood but not confident enough to help” as distinct reasons for escalation, and tagging each hold or escalation with its reason so the reasons can be priced differently. These are recorded in the discussion log with the change each one produced.

3 Agent design

The agent is defined by seven parts: what it observes, what it cannot see, the belief it holds over the hidden part, the actions available to it, the cost of a wrong action, the rule that picks an action, and the feedback it gets afterward.

Input. On each turn the agent receives the inbound message and the prior turns of the same conversation. It also receives a small amount of position context: the turn index and how many times a near-identical message has repeated. Roughly a third of the message types the agent must handle are position-dependent, so the same text can warrant a different action depending on where in the conversation it lands.

Hidden state. What the agent cannot see is split into two independent parts. The first is *readiness*, one of hot, warm, or cold, which is mutually exclusive: at any moment the lead really is in one of these. The second is whether the case *needs a human*, which is a separate property of the moment, not a fourth readiness level. The two are kept independent on purpose. If needs-human competed with hot, warm, and cold for the same probability mass, then evidence that a lead is hot would mechanically lower the estimated chance it needs a human, which is backwards: a hot, ready-to-buy lead is often exactly the one a person should handle. Section 6 reports that the failures are consistent with this independence rather than against it.

Belief. The belief is what the agent holds over the hidden state given everything seen so far: a distribution over hot, warm, cold that sums to one, and a separate probability that the case needs a human. The belief comes from a language model rather than a trained classifier, which makes calibration the load-bearing assumption of the whole design. A needs-human probability read too low makes the automated actions look cheaper than they are and suppresses escalation, and that is the direction the check finds. So the belief is checked against labeled cases before any threshold is trusted, and the check is reported rather than assumed.

Actions. The agent chooses one of five: answer, ask a qualifying question, hold, escalate-notify (tell a human while the conversation continues), and escalate-pause (stop and hand over). Escalation is two actions rather than one because notify and pause carry different costs: a needless pause can let a live lead go cold, while a needless notify only spends a person’s attention. Collapsing them would hide that difference from the decision rule.

Cost. A wrong action has a cost, and the costs are deliberately asymmetric. The ordering, from most to least expensive, is: a false assertion, a lead who needed a human but was held, a needless pause, a needless notify, a needless ask.

Correct actions cost nothing, except a correct pause, which carries a small residual so the policy does not treat pausing as free and over-use it. The false assertion is not merely the most expensive error; on the cases that forbid a direct answer, such as a request for legal or land documents, the answer action is removed from the choice set entirely. It is a hard constraint, not a high price, because there is no belief state that should make the agent send a false legal claim. The full matrix and the reasoning behind each value are in Section 4.

Policy. The policy computes the expected cost of each available action under the current belief and picks the lowest. It is myopic: it reasons one step ahead and does not plan over how the belief might change on future turns. This is enough for the answer, hold, and escalate actions, whose cost can be judged against the current belief. It is also enough for *ask*, though for a reason particular to this cost matrix rather than by construction: a question’s value is the belief it buys next turn less the fee it charges now, and here the fee alone is at least what acting now costs, at every belief. Section 6.3 proves the value of a question is negative at every belief on the unconstrained action menu and that the bound is horizon-independent, and Section 8 states what that does and does not license.

Feedback. After acting, the agent would in principle observe whether the lead converted or whether an escalated case was resolved. That signal is delayed, noisy, and partly counterfactual, and a missed escalation has no natural label at all: nobody records the lead that needed a human and did not get one. For the experiment in this paper, feedback is supplied by labeled synthetic cases instead, which is what makes recall on missed escalations measurable at all.

Human-reasoning function. The one human-like function the agent adds is to recognise when a decision is too costly to make alone and route it to a person, rather than always acting or always deferring. It maintains an explicit belief, updates it as evidence arrives, and escalates specifically when the expected cost of acting alone is high. This is what makes escalation a costed decision rather than a free safe harbour, and it is the behaviour the rest of the paper measures.

4 Probability model and decision rule

4.1 Belief

The agent never observes a lead’s true intent. It observes one message (optionally with a small amount of conversation context, the turn index and repeat count) and forms a belief over two hidden components.

The first is *readiness*, a distribution over three states:

$$b_r = (P(\text{hot}), P(\text{warm}), P(\text{cold})), \quad \sum_s P(s) = 1. \quad (1)$$

The second is *needs-human*, a single independent probability that the case requires a person regardless of how ready the lead is:

$$b_h = P(\text{needs_human} = \text{True}) \in [0, 1]. \quad (2)$$

These are kept separate on purpose. A lead can be ready to buy *and* need a human (someone has to confirm a booking),

Action	needs_human = False			needs_human = True		
	hot	warm	cold	hot	warm	cold
answer	0	0	0	10	10	10
ask	2	2	2	4	4	4
hold	6	1	0	8	4	3
escalate_notify	3	3	3	0	0	0
escalate_pause	6	5	5	2	1	1

Table 1: Cost matrix $C(a, s, h)$. Costs are expert-set; the ordering encodes that a missed escalation is far more expensive than a needless one. Correct actions cost 0; a correct escalate-pause carries a residual cost of 1 so the policy does not treat correct pausing as free. On `no_direct_answer` cases the *answer* action is removed from the feasible set, so its row does not apply.

or clearly not ready *and* still need a human (an abusive or competitor message a bot should not handle). A single confidence score cannot represent either case. Treating readiness and needs-human as independent is a modelling assumption, not a measured fact, and Section 6 does not validate it: the failures localise to one miscalibrated marginal, which is consistent with independence but not evidence for it, since the run never observes the joint.

Because the two are independent, the joint probability of any full state factors as a product:

$$P(s, h) = P(s) \cdot P(h), \quad (3)$$

where $s \in \{\text{hot}, \text{warm}, \text{cold}\}$ and $h \in \{\text{True}, \text{False}\}$. This gives six joint states. For the worked case in Section 4.3 the belief $b_r = (0.40, 0.50, 0.10)$ and $b_h = 0.30$ expand to $P(\text{hot}, F) = 0.28$, $P(\text{warm}, F) = 0.35$, $P(\text{cold}, F) = 0.07$, $P(\text{hot}, T) = 0.12$, $P(\text{warm}, T) = 0.15$, $P(\text{cold}, T) = 0.03$, which sum to 1.

4.2 Actions and costs

The agent chooses one of five actions: *answer*, *ask* (a qualifying question), *hold*, *escalate-notify* (tell a human while the conversation continues), and *escalate-pause* (stop and hand over). Each action has a cost in each of the six joint states, giving the 5×6 matrix $C(a, s, h)$ in Table 1.

The costs are expert-set, not learned. An earlier draft claimed their *ordering* was the load-bearing part and not the exact magnitudes; a sensitivity sweep supports that for one cell and refutes it for another, so the claim is stated more narrowly here. Sweeping the false-assertion cost from 7 to 30 preserves every ordering, yet the decisions shift midway: 16 misses and 43 escalations from 7 to 10 (cost 1.920 to 1.650), then 14 and 54 from 12 to 30 (1.590 to 1.520). Ordering-invariance holds locally, not across the range. The needless-notify cost is not like that. Sweeping it from 1 to 6, again preserving every ordering, moves the policy across almost its whole operating range: 1 miss and 96 escalations at a cost of 1, 8 and 73 at 2, 16 and 43 at 3, 23 and 28 at 4, and 25 and 23 at 5 or above. The reported operating point is a direct consequence of that one number’s magnitude, and it was set by judgment. Two cells are worth stating because they are where the two-part belief does its work. When a lead is hot *and* needs a human and the agent holds, the cost is 8, not

16: the two harms (the lead waits, no human arrives) overlap rather than stack, and pricing it at $6 + 10$ would exceed the false-assertion cost and break the framing in which a false assertion is the top of the scale. When that same lead is answered, the cost is 10 with no premium for hotness, since a false claim is equally false to a hot or a cold lead.

The false assertion is not merely priced high; it is a *hard constraint*. On the cases that carry a `no_direct_answer` flag, the *answer* action is removed from the feasible set rather than charged a high price. A price can be outweighed if enough other costs pile up; a constraint never can. There is no belief state that should ever make the agent send legal or land documents, so this is enforced as infeasibility, not as a large number.

4.3 Decision rule

The policy is *myopic and one-step*: it picks the action with the lowest expected cost under the current belief, with no lookahead over future turns. For a feasible action a ,

$$\mathbb{E}[C(a)] = \sum_s \sum_h P(s) P(h) C(a, s, h), \quad (4)$$

and the chosen action is

$$a^* = \arg \min_{a \in \mathcal{A}_{\text{feasible}}} \mathbb{E}[C(a)]. \quad (5)$$

For the worked case above this yields expected costs of escalate-notify 2.10, ask 2.60, answer 3.00, hold 3.68, escalate-pause 4.20, so the agent escalates, and the realised cost is 0 because the case truly needed a human.

4.4 Where the threshold comes from

Because the readiness rows for *answer* and *escalate-notify* are flat across readiness (a false answer costs 10 in any readiness state where a human is needed; a needless notify costs 3 in any state where one is not), the choice between them reduces to a single threshold on b_h . Answer is cheaper while $10 b_h < 3(1 - b_h)$, and escalate-notify takes over when the inequality flips, at

$$b_h^* = \frac{3}{13} \approx 0.2308. \quad (6)$$

This crossover is independent of the readiness distribution, precisely because both rows are flat across readiness. In the worked case $b_h = 0.30$ clears it by 0.069, which is why the agent escalates even though the belief is only modestly above the line.

The precision of the fraction is misleading, however, and it is worth saying so before the results lean on it. The elicited b_h takes only eight distinct values over the 100 cases, all at one decimal place, and none of them falls in the open interval $(0.2, 0.3)$. Every threshold in the half-open $(0.2, 0.3]$ therefore produces identical decisions on every case — the value gap is open at the top because 17 of the 100 cases sit at exactly 0.3, while a threshold at 0.3 still leaves all 17 on the escalate side — and $3/13 \approx 0.231$ is one arbitrary point inside a band the measurement cannot resolve. Nothing in the results distinguishes it from 0.25.

This clean single threshold holds only when the runner-up action is *answer*. When *hold* is the competing action its row is

not flat across readiness, so the threshold becomes readiness-dependent, and the spread is much wider than the two hold cases in the failure analysis (which flip at 0.306 and 0.373) would suggest. Solving $E[\text{notify}] = E[\text{hold}]$ at each case’s actual readiness distribution puts the crossover anywhere in $[0.018, 0.500]$ on the 73 cases where it falls in $[0, 1]$ at all; on the other 27 it is negative, meaning *hold* never wins for any admissible b_h . At the pure readiness states the crossover is -0.600 (hot), 0.333 (warm) and 0.500 (cold). So $3/13$ is the threshold for one specific pairwise comparison, not a single operating point for the policy. The policy is still a single arg min; it is only the *reduced* one-dimensional threshold that shifts with which action is the alternative, and quoting $3/13$ alone understates how much of the decision boundary depends on readiness.

4.5 Baseline

The baseline is the same decision rule run over an identical belief with a *uniform* cost matrix, every non-zero cost replaced by one. This isolates exactly what cost-awareness buys: the belief, the feasible-set constraint, and the expected-cost machinery are all held fixed, and only the asymmetry is removed.

5 Test method

5.1 Cases

The agent is tested on 100 synthetic cases. Each case is one inbound message with a label for the true readiness (hot, warm, cold) and whether it truly needs a human, plus the position context (turn index and repeat count) the belief sees. The cases are built from eleven archetypes that between them cover the situations where the cost trade-off actually bites: template openers, requests for photos, one-word messages, ready-to-book buyers, requests for legal or land documents, messages that suspect the agent is a bot, competitors fishing for pricing, captionless media, over-sharers, polite time-wasters, and abusive or off-topic messages. The distribution is deliberately weighted toward the archetypes where a wrong action is expensive rather than made uniform, and it is skewed toward harder cases: 42 of the 100 truly need a human, chosen so that recall on missed escalations is measurable at all. How far that is above a live inbox is not known here — no measured base rate for this channel is available to the author — so the 42% should be read as a design choice made for measurability, not as a calibrated departure from a known population.

The readiness distribution of the case set is also worth stating explicitly, because it does not match the prior the belief module itself carries. That prior is (hot, warm, cold) = $(0.02, 0.13, 0.85)$, whereas the true labels in the case set are 23 hot, 38 warm and 39 cold. The set is not a sample from the population the design assumes; Section 6 reports what happens when the results are reweighted toward that prior.

The labels are the author’s judgment of the intended state, not outcomes observed in production. This is stated plainly as a limitation in Section 8: the readiness and needs-human labels are what the message was written to represent, not what a real lead later did.

5.2 Beliefs

For each case the belief is produced once by a language model (gpt-4o-mini) and cached to disk, keyed by case id. Both policies are then run over the identical cached beliefs, so any difference between them comes from the decision rule and not from variation in the belief. Caching also makes the run reproducible: the language model is non-deterministic, so re-generating the beliefs would move every reported number, and the cache is committed alongside the code for that reason.

5.3 Split

The 100 cases are split 50/50 into a development and a test half with a fixed seed, stratified by archetype and sub-variant so that every variant appears in both halves and the needs-human count is balanced (21 in each). The split is not redrawn per run. Because the halves are matched by construction, close agreement between development and test totals reflects that construction rather than out-of-sample generalization, and Section 6 reads the split numbers with that in mind.

The same split does double duty. The calibration map of Section 6.9 is fitted on the 50 development cases and every number that map is judged by is scored on the 50 test cases, so the calibrated beliefs on the development half are in-sample and are not reported as a result. Matching the halves by construction cuts both ways here. It is why close agreement between the two is not evidence of generalization, and it is also why they are an acceptable place to fit a one-dimensional map and then score it: the fit cannot be flattered by an easier half to score on, because the archetypes and the needs-human count are balanced across the two.

5.4 Policies and baselines

Two policies are compared over the same beliefs. The *cost-aware* policy is the minimum-expected-cost rule of Section 4 under the full asymmetric cost matrix. The *uniform* baseline is the same rule with every non-zero cost replaced by one, which holds the belief, the feasible-set constraint, and the expected-cost machinery fixed and removes only the asymmetry. This isolates what cost-awareness buys. Three fixed-action baselines are also reported for reference: always answer, always ask, and always escalate-notify.

One implementation detail changes a reported number, so it is disclosed here rather than left to the failure analysis. Exact ties in expected cost were originally resolved in the order the five actions are declared, which resolves toward *answer* — the action with the largest worst-case cost, and the one a policy premised on the expense of false assertion should reach for last. Ties are now resolved safest-first, in an order derived from the cost matrix itself by worst-case cost (*tie_break_order*) rather than hand-written. The run reported below deliberately retains the original order, via the `--legacy-tie-break` flag, so the committed result artifact stays reproducible action for action. Under the corrected default exactly one of the 100 decisions changes and the mean cost falls from 1.72 to 1.65; Section 7 gives the case and the arithmetic.

5.5 Metrics

Accuracy alone would hide the thing the design is about, so the run reports mean realised cost per case, the number of missed escalations (a case that needed a human but did not get one), escalation precision and recall, the number of escalations, the number of hard-constraint violations, and the calibration of the belief. Accuracy is reported too, in Table 2, precisely so that a reader can watch it hide that thing rather than take the claim on trust: it is a count of decisions that match a label, it is blind to which mistake was made, and under the asymmetric matrix the two are not the same question. Calibration is measured as expected calibration error on the needs-human marginal, since that is the probability the escalation threshold acts on. Two further measures are reported for the held-out fit of Section 6.9, both fixed before it and both on the same marginal: cross-entropy and Brier score.

Cost, split two ways. Mean realised cost is reported as two columns rather than one, because exactly one of the five actions in Table 1 buys information and the other four do not. What *ask* costs is the price of a question, and it leaves the case where it found it; what the other four cost is the price of having acted. Splitting them keeps a policy that pays only for questions from looking like a policy that acted cheaply, which is what a single total does to the *always-ask* row. The split is zero for every policy that never asks, which in Table 2 is every policy but that one.

Entropy thresholds. One further quantity is measured in bits rather than in cost points, and it is defined here rather than where it is used, because a later section places a threshold on it. $H(b)$ is the entropy of the belief over the six hidden states, so it runs from 0 to $\log_2 6 \approx 2.585$ bits. The entropy-threshold rule swept in Section 6.6 asks whenever $H(b) > \tau$ and otherwise takes the cheapest action other than asking, which on these beliefs is the action the policy of Section 4 takes case for case. That makes τ the only thing $H(b)$ is ever compared against. No quantity in bits is compared to a quantity in cost points anywhere in this paper: the entropy selects which cases are asked about, and the cost matrix scores what happens.

τ is not set in absolute bits. It is a grid of eleven thresholds placed at the quantiles 0.0, 0.1, ..., 1.0 of the observed $H(b)$ on the arm being scored, linearly interpolated. The reason is that the observed distribution is bunched: on the run's own beliefs $H(b)$ spans 0 to 2.456 bits with a median of 2.017, so an evenly spaced grid over the theoretical range would put most of its points in a region holding almost no cases, and would report the spacing of the grid rather than the behaviour of the rule. Deciles make the step commensurate with the observed spread by construction. $H(b)$ is rounded to twelve decimals before it meets τ , a figure derived from the measured float-noise bound on each arm rather than chosen.

Three things follow from that definition rather than from any result, and are stated here so that they are not read as findings later. A decile grid on a tied distribution repeats values: the eleven quantiles give eight distinct thresholds on the run's own beliefs and eleven on the re-elicited beliefs of Section 6.9, and two quantiles that yield the same τ yield the same set of asked cases and the same cost by construction

rather than by coincidence. The same quantile is a different number of bits on each set of beliefs, because $H(b)$ moves when the belief moves, so the arms are compared at equal quantiles and never at equal bits. And the grid is taken over all 100 cases of an arm while the sweep is scored on the 50 test cases, so the top decile asks about nothing at all whenever an arm's highest-entropy case is a development case.

6 Results

Table 2 reports all five policies over the 100 cases, on the shared cached beliefs.

6.1 What cost-awareness buys

The cost-aware policy reaches a mean cost of 1.72 against 2.58 for the uniform baseline, a reduction of a third over 100 shared beliefs. That reduction is against the uniform-cost baseline and not against *always-notify*, which Section 6.7 reports as a near-tie on the same 100 cases; the two must not be read for each other. Because the two policies compared here differ only in whether the cost matrix is asymmetric, this gap is what cost-awareness buys and nothing else. The two disagree on 46 of the 100 cases, and the cost-aware policy is not cheaper on more of them: it is cheaper on 15, the baseline is cheaper on 18, and 13 are ties. It wins overall because its losses are small and its wins are large. Almost every swap it makes is answer to hold or answer to escalate-notify, trading many small costs of two or three for a few avoided false assertions of ten. The point of the design is not to be right more often but to be wrong more cheaply.

The accuracy column is where that stops being a slogan. The baseline is the more accurate of the two policies, 0.700 against 0.670, while costing 2.58 against 1.72: the ordering by accuracy is the reverse of the ordering by cost, on the same 100 beliefs, with the same notion of which action is correct in each state. Nor is the inversion confined to the two. *always-answer* reaches 0.660, within 0.010 of the shipped policy, at roughly double its decision cost — and every one of its 34 errors is a false assertion at the matrix's top price, so its entire cost is 34 mistakes of one kind. Accuracy cannot see that, because it counts decisions and the matrix prices them. Ranked by the accuracy column, the baseline sits above the shipped policy and a policy that never escalates sits third; the row that column ranks last, *always-notify*, is the second cheapest in the table. So the column does not order these five policies the way the objective does, at either end of it.

6.2 What the baseline actually is

One further check weakens how much that gap can be attributed to the machinery. The uniform baseline selects the same action as a plain threshold of 0.5 on b_h — ignoring readiness, the cost matrix, and the expected-cost computation entirely — on 100 of the 100 cases. Its behaviour is fully described by one number. The cost-aware policy is not reducible in the same way, because *hold* is chosen on 27 cases and that choice does depend on readiness, but the comparison in Table 2 is between a readiness-sensitive rule and a 0.5 threshold, and part of the reported gap is the distance from 0.5 to 3/13 rather than the asymmetry as such.

Policy	Accuracy	Precision	Recall	Missed	Decision cost	Info cost	Human %
cost-aware	0.670	0.605	0.619	16	1.72	0.00	43
uniform baseline	0.700	0.750	0.429	24	2.58	0.00	24
always-notify	0.420	0.420	1.000	0	1.74	0.00	100
always-ask	0.580	—	0.000	42	0.00	2.84	0
always-answer	0.660	1.000	0.190	34	3.40	0.00	8

Table 2: All five policies over the 100 cases. *Accuracy* is the fraction of cases where the escalate / do-not-escalate decision matches the needs-human label; precision and recall are of escalation; *Missed* counts cases that needed a human and did not get one. *Info cost* is the part of mean realised cost paid by *ask* and *Decision cost* is the rest, so the two sum to the single mean cost reported before and the table carries the split rather than the total. *Human %* is the percentage of cases routed to a person, which over 100 cases is numerically the escalation count; `experiments/table_metrics.py` asserts that equality per policy rather than assuming it, recomputes the four pre-existing columns against `results/run.json` and finds no disagreement, and writes `results/table-metrics.json`. Lower cost and fewer misses are better; accuracy is reported as the foil the Metrics subsection above says it is, not as an objective. Two of its entries are forced rather than measured. *always-notify* escalates on every case, so its accuracy is its own precision and both are the base rate of the label, 0.420; *always-ask* escalates on none, so its accuracy is one minus that base rate, which is the majority-class score — the table therefore contains its own trivial-classifier reference point, and no policy in it beats that score by more than 0.120. The cost-aware row is the legacy tie-break configuration described in Section 5; under the corrected default its decision cost is 1.65 and every one of the other six entries is unchanged, which is checked column by column rather than asserted. No policy records a hard-constraint violation, including the fixed-action baselines — the constraint forbids only *answer*, and on all 8 restricted cases the cost-aware policy escalates on cost grounds before feasibility is consulted, so the constraint never changed a decision. Zero violations here is therefore not evidence that the constraint mechanism works; it is untested. *always-answer* shows precision 1.000 on 8 escalations, which is an artifact: the policy never escalates, and those 8 are the restricted cases where the constraint removed *answer*, leaving the fixed-action policy to fall back to escalate-notify by construction rather than by any comparison of costs. All 8 happen to need a human, so the precision is 8/8 on a denominator the policy did not choose. It should not be read as the baseline discriminating well.

That agreement is not a property of these cases, and the reason is worth setting out, because it changes what the baseline is a baseline for. Flattening every non-zero entry to one — which is how the uniform matrix C_u is defined, at `src/costs.py:109–112`, derived from Table 1 rather than written by hand — does not produce a weaker cost model. It produces no cost model at all, and what the expected-cost rule then computes has a closed form:

$$\begin{aligned}
\mathbb{E}_b[C_u(\text{answer})] &= b_h, \\
\mathbb{E}_b[C_u(\text{escalate-notify})] &= 1 - b_h, \\
\mathbb{E}_b[C_u(\text{ask})] &= 1, \\
\mathbb{E}_b[C_u(\text{escalate-pause})] &= 1, \\
\mathbb{E}_b[C_u(\text{hold})] &= 1 - P(\text{cold})(1 - b_h).
\end{aligned} \tag{7}$$

Three consequences follow, and none of them mentions the data. *Answer* is preferred to *escalate-notify* exactly where b_h falls below the threshold named above, so that threshold is the argmax of the needs-human marginal and not a value anyone selected. *Hold* can never strictly win, since the first and last lines of (7) differ by $(1 - b_h)(1 - P(\text{cold}))$, which is never negative — so readiness, which the full matrix consults to choose *hold* on the cases counted above, has no purchase here even though C_u still contains it. And the two middle lines sit at the flattened maximum, which the smaller of the first two is always strictly below, so on the unconstrained menu neither *ask* nor *escalate-pause* is ever selected. That last conclusion is reached without Theorem 1 or any part of its argument: the baseline’s silence about asking is arithmetic in a matrix with one magnitude, whereas the theorem is a statement about the practitioner’s magnitudes. Removing *answer* leaves one corner where it would not hold — at $b_h = 0$ with $P(\text{cold}) = 0$ the four remaining actions all cost the same, and the degener-

ate ordering described in Section 7 would resolve that tie to *ask* — but no restricted case in the reported set sits there.

So the agreement above measures something that could not have come out otherwise. The identity holds at every belief in the readiness simplex crossed with $b_h \in [0, 1]$, so no case set could have contradicted it, and the check is executed rather than asserted: `experiments/policy_ladder.py` compares all five closed forms against the implementation on every committed belief and all five actions, and `results/policy-ladder.json` records the largest disagreement, which is at floating-point rounding. What this settles is the shape of the comparison in Table 2. Those two policies are not a cost model beside a blunter cost model; they are a rule that prices the belief’s spread asymmetrically beside a rule that reads nothing but the mode of the belief’s needs-human half. Flattening removes the asymmetry and the use of the spread together, which is why the reported gap cannot be split between them by this comparison alone — and why the rung between the two, a rule that holds the practitioner’s magnitudes but reads only the mode, has to be built separately rather than recovered from Table 2.

A threshold sweep separates the two contributions, and it does not go cleanly. No threshold on b_h reproduces the cost-aware policy’s 43 escalations: because the elicited values are quantized, the sweep jumps from 29 escalations at $t = 0.35$ (23 misses, cost 2.600) to 46 at $t = 0.25$ (13 misses, cost 1.810), both counted over the same 100 cases as Table 2. The cost-aware policy’s 16 misses at 43 escalations, also over all 100, is therefore not straightforwardly better than a tuned threshold at matched load, because matched load is not attainable; the nearest bracketing thresholds are worse on one axis and better on the other. Tuning a threshold on the 50 development cases by realised cost selects $t = 0.00$ — es-

calate everything — at a cost of 1.740, which is its cost on the development half, on the test half and over all 100 alike; it recovers the always-notify result rather than the cost-aware one. That selection is specific to the half it was tuned on: on the test half $t = 0.05$ and $t = 0.10$ both cost 1.68 against $t = 0.00$'s 1.740, so tuning there by the same criterion would not have chosen escalate-everything.

6.3 Two actions that are never used, and why neither can be

Of the five actions, the cost-aware policy selects only three across all 100 cases: *answer* 30 times, *hold* 27, *escalate-notify* 43. *Ask* and *escalate-pause* are selected zero times, and the same census is reproduced independently by the value-of-information analysis below. This is a property of the matrix rather than of the cases, and this section proves it rather than observing it. Each of the two unused actions fails for a different reason, and neither reason mentions the data.

Asking. *Ask* costs 2 when no human is needed against 0 for *answer*, and 4 when one is needed against 0 for *escalate-notify*, so it is dominated at both ends. That arithmetic does not settle the middle, and the middle is what has to be settled, because a question is not priced by its own row alone but by its row plus the cost of acting afterwards. Write

$$V_{\text{act}}(b) = \min_{a \neq \text{ask}} \mathbb{E}_b[C(a)] \quad (8)$$

for the cost of acting now on the best non-ask action, let $c_F = 2$ and $c_T = 4$ be the *ask* row, and let $\alpha = 10$ and $\nu = 3$ be the cost of a false answer and of a needless notify. Asking question q pays only if $\mathbb{E}_b[C(\text{ask})] + \sum_u P_b(u) V_{\text{act}}(b^u)$ falls below $V_{\text{act}}(b)$, where b^u is the posterior after answer u . Every cost in Table 1 is non-negative, so that continuation term is non-negative, and a necessary condition for a positive value of information is that $\mathbb{E}_b[C(\text{ask})] < V_{\text{act}}(b)$ at some belief. No belief satisfies it.

Theorem 1. *With the costs of Table 1, at every belief b in the readiness simplex crossed with $b_h \in [0, 1]$,*

$$V_{\text{act}}(b) - \mathbb{E}_b[C(\text{ask})] \leq -\frac{2}{13} \approx -0.1538,$$

and the bound is attained. The value of information of every question at every belief is therefore at most $-2/13$, and ask is never the arg min on the unconstrained action menu.

Proof sketch. $\mathbb{E}_b[C(\text{ask})] = c_F + (c_T - c_F) b_h$ is affine in b_h and free of readiness, and $V_{\text{act}}(b) \leq \min\{\alpha b_h, \nu(1 - b_h)\}$ because the *answer* and *escalate-notify* rows are themselves flat across readiness. The difference is thus bounded by a piecewise linear function whose *answer* branch has slope $\alpha - (c_T - c_F) > 0$ and whose *escalate-notify* branch has slope $-(\nu + c_T - c_F) < 0$, both signs following from $-\nu < c_T - c_F < \alpha$, which the implementation asserts rather than assumes. Its maximum is at the crossing $b_h = b_h^* = 3/13$ and equals $[\alpha\nu - (\alpha c_F + \nu c_T)]/(\alpha + \nu) = (30 - 32)/13$. The bound is attained where V_{act} meets that envelope: at the all-hot vertex with $b_h = 3/13$, *hold* costs $84/13$ and *escalate-pause* $66/13$, so neither lowers the cap, and $V_{\text{act}} = 30/13$ against $\mathbb{E}_b[C(\text{ask})] = 32/13$. The computation is in exact rational arithmetic; a deliberately crude 60-step float grid over

the same domain reaches -0.166667 and never exceeds the closed form. $\square$

The maximum is taken over the whole belief space and not over the 100 elicited beliefs, which is what exposes the two errors in the claim this passage replaces. There is no narrow middle band on the unconstrained menu: the ceiling is negative at every belief in the simplex, and its largest value anywhere is $-2/13$. And quantization is not the cause. The elicited b_h takes eight distinct values, but a re-elicited arm carrying 100 distinct values leaves the region exactly as empty: across four belief arms and 400 case-arm pairs, no case has a positive ceiling, and the per-case maximum on the published beliefs is -0.4000 . A finer belief does not find the band, because there is no band to find.

Attainment is not a second result but what makes the first one sharp. A negative ceiling could otherwise be an artefact of a loose bound, and the strongest form of that objection is answerable directly. On 16 of the 400 case-question pairs scored on the published beliefs the continuation term $\sum_u P_b(u) V_{\text{act}}(b^u)$ is exactly zero: the question resolves the state and the action taken afterwards is free. Asking still loses, and it loses by the full ceiling at that belief rather than by a residue, because the fee is paid before the answer arrives. The count is 52 such pairs on the calibrated arm, 12 on the re-baselined arm and 0 on the raw arm. The theorem does not rest on those pairs; they are where its inequality stops being one.

Nor does the result rest on the policy's one-step horizon. Define the cost of acting under a budget of k remaining questions by $W_0 = V_{\text{act}}$ and

$$W_k(b) = \min \left\{ V_{\text{act}}(b), \mathbb{E}_b[C(\text{ask})] + \sum_u P_b(u) W_{k-1}(b^u) \right\}, \quad (9)$$

so that the policy reported here is W_1 . Two facts already in hand close every k at once. Theorem 1 is quantified over all beliefs, and posteriors are beliefs: b^u , and every deeper node of any lookahead tree, is a point of the same simplex, so the ceiling binds at every node and not only at the root. And $W_k \geq 0$ for every k , by induction from the non-negativity of Table 1. Together, at every belief and every $k \geq 1$, the ask branch costs at least $\mathbb{E}_b[C(\text{ask})] \geq V_{\text{act}}(b) + 2/13$, so $W_k = V_{\text{act}}$ and the tree collapses to acting now. Every term dropped in that chain is non-negative, so deeper lookahead can only make the ask branch dearer: $2/13$ is a floor on the gap, not an estimate of it. The second fact is trivial, following from the cost matrix alone with no reference to the beliefs, the answer model or the question set, and that is exactly why it can be asserted at every node of a tree nobody enumerates.

The theorem is a statement about four numbers, and the part that transfers is worth separating from the part that is ours. Rearranged, a question can be rational at some belief on the unconstrained menu only if

$$\frac{c_F}{\nu} + \frac{c_T}{\alpha} < 1, \quad (10)$$

that is, only if the fee for asking when no human is needed, measured against the cost of a needless escalation, plus the fee for asking when one is needed, measured against the cost

of a false answer, together come to less than the whole. Here the sum is $2/3 + 4/10 = 16/15 > 1$, so the condition fails, and it fails by 6.25%: scaling the *ask* row by $\lambda = 15/16$, to (1.875, 3.75), puts the ceiling exactly at zero. That is the useful form of the result. A designer with a cheaper question — a single-tap reply, an inference from metadata — crosses the line, and the same inequality says where the line is. What this experiment shows is not that asking is useless in general, but that on this matrix asking is priced out, by a margin small enough that the pricing rather than the principle is what decides it.

One qualification is load-bearing, and the theorem is stated with it: the menu is the unconstrained one. On `no_direct_answer` cases *answer* is removed from the feasible set, and with the cheapest row gone the ceiling turns positive — +1 at the all-hot vertex with $b_h = 0$. Along the ray through that maximum the positive region is $b_h < 1/5$, bounded there by *escalate-notify* rather than by *hold* or *escalate-pause*. So a region in which asking is rational does exist, and it is not reached here: 8 of the 100 cases carry the constraint, every one of them at $b_h \geq 0.40$, and none lies inside the region. The two ways of being empty are different, and they are not merged. On the calibrated arm the emptiness is structural: the fitted map’s reachable-score floor sits above $1/5$, so no input can produce a belief low enough to qualify — the same floor reported with the calibration result in Section 6.9. On the published, rebaselined and raw arms it is contingent: 31 of the raw beliefs do fall below $1/5$, and the region stays empty only because none of those cases carries `no_direct_answer`. That is a fact about this dataset, not about the matrix.

Pausing. *Escalate-pause* is dominated by *escalate-notify* in every column of Table 1, and dominance in the matrix is a stronger statement than absence from the census. The gap is 3 in the hot state where no human is needed, 2 in the warm and cold states where none is needed, and 2, 1 and 1 in the three states where one is: strictly positive in all six, and between 1 and 3. Expected cost is a belief-weighted average over columns, so no belief can invert a strict column-wise ordering, and *escalate-pause* cannot be the *arg min* anywhere — not merely at the 100 beliefs elicited here.

That also bounds abstention as a design option. A rule that hands the conversation over rather than notifying converts *escalate-notify* decisions into *escalate-pause* decisions, so it raises realised cost by between 1 and 3 on every case it touches, and it cannot reduce the count of missed escalations at all, since a miss is a case needing a human on which the agent did not escalate and both escalate actions count as escalating. Only a rule that converts a non-escalation into a pause can move that count. Abstention is therefore priced rather than unexplored, and what remains empirical is how many cases such a rule touches and whether the misses it avoids are worth the penalty it pays.

The census is unchanged: the paper describes a five-action set and every reported number comes from a three-action policy. What changes is the reason. The two unused actions are not untested by this experiment — both are tested, and each fails for a different proven reason: *ask* by a ceiling on the

value of information that is negative at every belief on the unconstrained menu, *escalate-pause* by column-wise dominance in the matrix. The one-step horizon is a real limitation of the policy and is stated as one in Section 8, but it is not the reason *ask* goes unused and it does not compound this: the horizon was the candidate explanation, and Theorem 1 together with the collapse of W_k removes it. The action most obviously suited to an ambiguous inbound message is unused because it is priced out, not because it is unpriced.

6.4 Running the lookahead that the theorem collapses

Theorem 1 and the collapse of W_k are proofs about a matrix, and neither was ever executed. W_1 appears above as an inequality that a policy would have to satisfy; no policy in Table 2 evaluates it, because the shipped rule compares the *ask* row against the other rows and stops there. So W_1 is built here and run: it prices a question as the fee plus the expected cost of acting on the actual posterior, answer by answer, over the four questions of the committed answer model. Alongside it runs a second arm that takes the continuation term to be zero everywhere — not a policy anyone would ship, but exactly the necessary condition stated before the theorem, and therefore the largest λ at which any question whatsoever could be worth its price. Both arms decide on the same committed beliefs, in exact rational arithmetic; they live in `experiments/voi_boundary.py` and every number here is recorded in `results/voi-boundary.json`.

At the practitioner’s prices both arms ask on zero of the 100 cases, and W_1 plays the shipped policy’s action on all 100 under the corrected ordering of Section 7 — the single disagreement with the committed decisions is that section’s one tie-break case and nothing else, which the arm checks rather than assumes. The collapse is therefore measured and not only proved, and the census in this section is reproduced by a policy that had the option of contradicting it.

Where the switch actually lands. Scaling the *ask* row by λ turns the census into a boundary question, and the boundary can be solved for rather than searched: under W_1 a case asks exactly when λ falls below $[V_{\text{act}}(b) - \sum_u P_b(u)V_{\text{act}}(b^u)]/\mathbb{E}_b[C(\text{ask})]$, and under the other arm below the same expression with the continuation term struck out — one rational per case and question either way, so each switch is the largest of those and not a grid artifact. A crude sweep in twentieths is run anyway and agrees with the closed form at every step, which is the pattern the proof above already uses. The necessary-condition arm switches at $5/6$. W_1 switches at $11/30$, on `a01-first-003`, with the question aimed at b_h .

Neither is the repricing derived above. The necessary-condition arm falls $5/48$ short of it, and that gap is the belief set rather than the algebra. The ratio it maximises rises into the crossing b_h^* and falls out of it, so the best available belief is whichever elicited value sits nearest; the two that bracket b_h^* give $5/6$ and $21/26$. Pushing the theorem’s own witness belief through this arm’s expected-cost code returns the repriced λ exactly, so the arithmetic is not where the shortfall comes from. This does not soften the claim above that quantization

is not the cause: that claim is about the sign of the ceiling, and it stands — a finer belief grid finds no positive region because there is none to find. What quantization sets is the margin, how close the best elicited belief gets to a break-even the whole simplex attains. A result about a sign and a result about a distance are not the same result, and the second does not qualify the first.

What the answer model adds, and what it costs. The distance from 11/30 to 5/6 is the distance between the theorem’s bound and the questions that exist, and separating them separates what rests on the answer model from what does not. The necessary-condition arm never consults it, so the limitation recorded in Section 8 does not reach 5/6; 11/30 is inside that limitation’s scope and is reported as a property of this question set rather than of questions in general. Two features of the measurement say the machinery is not manufacturing value. The control question, whose answer cannot depend on the state, returns a break-even of exactly zero on every case — it is never worth buying at any price above free. And the question that attains 11/30 is the one aimed at b_h , which is the quantity the act comparison turns on, while the one aimed at readiness and the one aimed at both halves come in lower; the ordering is the one the matrix implies, not one the implementation chose.

One methodological point is load-bearing rather than incidental. The question aimed at both halves of the state couples them, so its posterior is not a product of its own marginals — on 96 of the 100 cases it is not — and the belief representation of Section 4, a readiness marginal plus a scalar, cannot express it. Expected cost is therefore computed against the full six-state distribution in this arm, and checked against the paper’s own expected-cost function on the 100 priors, where both apply and must agree. Projecting those posteriors onto their marginals instead would have understated what an answer is worth, in the arm whose whole purpose is to find the price at which it becomes worth something.

Asking that the belief endorses and the outcome does not. Noticed after the sweep ran, not predicted, and it is an observation about 100 labelled cases rather than about the rule. Below 11/30, W_1 begins to ask, and for a stretch its realised mean cost is *above* what it pays by never asking — the figure Table 2’s footnote reports for the corrected tie-break — peaking at 1.828 around $\lambda = 1/4$ before falling below it somewhere between 3/20 and 1/10, which the sweep brackets rather than locates. The lookahead is an expectation under the belief; where the belief is wrong the question is worth buying and wasted in outcome, and the two facts are consistent. What it warns against is reading the first λ at which a policy asks as the first λ at which asking helps.

What was predicted, and one miss. This arm’s pre-registration was a prediction rather than a restatement: the switch points were not known when it was written. Nine of its ten fields matched, including 5/6, the 5/48 shortfall, its attribution to the belief grid, the control question’s zero and which question would attain the real switch. One did not. W_1 ’s switch was predicted below 1/4, on the reasoning that a single answer rarely moves the act decision, and it landed at 11/30: the question aimed at b_h flips the decision on two

of its three answers at the maximising case, which is more than one question was expected to buy. That field is recorded as a miss in `results/voi-boundary.json` rather than adjusted, and the arm’s exit status treats a divergence there as a result to report and not a condition to fail.

6.5 What the magnitudes buy at a point estimate

Between a fixed action and the policy of Section 5 sits a rung that keeps the practitioner’s magnitudes and throws away the belief’s spread: collapse the belief to its single most likely joint state and play the cheapest action for that state under Table 1. Because the two halves of the belief are independent by construction, the joint mode is just the pair of marginal modes, so nothing has to be enumerated to find it. This is certainty equivalence, and the state-to-action map it induces is not written down anywhere — it is the column-wise argmin of the matrix, so changing any cost moves the map with it. The arm decides on the same committed beliefs as Table 2 and has its mistakes priced by the same matrix; it lives in `experiments/policy_ladder.py`, and every number in this subsection is recorded in `results/policy-ladder.json`.

What it shows is that at a point estimate the practitioner’s magnitudes buy almost nothing over the flattened ones. The arm departs from the uniform baseline on 20 cases, and on 19 of them the two candidate actions cost *the same* in the column the decision turns on: *answer* and *hold* are both 0 against (cold, F). The matrix expresses no preference there, so what selects *hold* on those 19 is the tie-break convention and not any number the practitioner set. Those 19 swaps save 44 realised points. The twentieth case is the only one where the magnitudes do prefer the arm’s action at the mode, and there the arm is wrong and pays 4. The net margin is 40 points, and it belongs to a convention rather than to a cost model.

Removing the convention removes the margin. Under the ordering that Section 7 describes as the original — indifference resolved toward *answer* — the arm’s *hold* column does not shrink but disappears, all 19 decisions becoming *answer*, and the mean rises from 2.18 to 2.62. That is worse than the uniform baseline in Table 2, and it has to be: in that configuration the two policies choose identically everywhere except the one case the magnitudes genuinely drive, so the arm’s total is the baseline’s total plus that case’s 4 points. Sharper still, only 9 of the 19 indifferent swaps move realised cost at all — for the other 10, answering would have been just as right. A rung whose whole advantage is nine cases decided by a convention, in a column where the matrix says nothing, is not the rung it looks like.

The mechanism is that collapsing the belief manufactures the ties it then has to break. Under the full belief an exact tie in expected cost requires a coincidence across six weighted entries, and it happens on one case out of the hundred, as Section 7 reports. At a point estimate the comparison is between two raw entries of a single column, and Table 1 contains an exact tie in the (cold, F) column, so every case whose mode lands there inherits it — nineteen of them. Certainty equivalence promotes an implementation detail into the arm’s dominant mechanism, and it does so by discarding exactly

the information that would have broken the tie. This is also why the gap reported in Section 6 cannot be decomposed by inserting this rung between its two ends. The asymmetry only ever acts through the spread, so collapsing to the mode does not weaken the asymmetry’s contribution while leaving it measurable; it removes the asymmetry’s purchase altogether. There is no stable midpoint here to read a split off of, and the arm’s position between the two reported means is an artifact of which tie-break it runs under.

The one *ask*, against the ceiling. This arm selects *ask* once, on `a05-restricted-043`, and it is the only place in this paper where a comparison of costs selects that action — the fixed rule in Table 2 asks by construction, and the rule in Section 6.6 asks on an uncertainty trigger that consults no cost at all. It does not bear on Theorem 1, because the two are statements about different objects. The theorem compares the price of a question against acting now *and* against acting once the answer is in, and it holds at every belief. This arm computes neither term. It reads one column of Table 1 at a state it has chosen to treat as certain, finds *ask* at 2 below *escalate-notify* at 3 once the constraint has removed *answer*, and plays it. A cost-column arg min at a belief that has already thrown away its own uncertainty is a degenerate table lookup, not a value-of-information claim, and the realised cost marks the difference: the case did need a human, and the action that the theorem’s own accounting would have taken was free.

What was checked, and what was noticed. The two halves of this subsection have different standing, and neither lends any to the other. The arm’s own numbers were pre-registered — the state-to-action map, the mean, the miss count, the escalation count, precision and recall, all written down before the code existed — and the measured output matched every field. That match confirms an implementation and establishes nothing about the world: the arm is a deterministic function of beliefs that were already committed, so it could not have discovered anything, and a divergence would have indicted the specification rather than the data. The attribution is the opposite case. It was found post-hoc, while checking whether the arm was comparable to a table generated under the other tie-break order, and `results/policy-ladder.json` labels it as such at the site. Its arithmetic was written down before it was implemented and matched too, which is a check on the arithmetic and credits nothing further.

6.6 Asking when uncertain, priced

Theorem 1 says asking is never rational on the unconstrained action menu. It does not say what the rule a reader reaches for instead would cost, and that rule is not a cost rule at all: ask whenever the belief is uncertain, on the intuition that uncertainty is what a question is for. Pricing it is not a search for a better baseline to beat. It is the same result measured instead of proved — if the ceiling holds, every setting of the rule has to lose, and the only open question is by how much.

The rule asks whenever $H(b) > \tau$ and otherwise takes the V_{act} argmin, which reproduces the shipped policy’s action case for case, so it differs from the shipped policy on exactly the cases where it fires. It is scored on the 50 test cases over

q	reported		rebaselined		raw		calibrated	
	ask	+cost	ask	+cost	ask	+cost	ask	+cost
0.0	50	113.62	50	109.64	50	128.02	50	118.98
0.1	37	83.12	39	73.12	47	118.02	44	94.98
0.2	37	83.12	39	73.12	40	89.50	42	86.50
0.3	33	73.20	34	67.20	36	84.54	35	70.90
0.4	14	40.60	15	34.60	31	70.98	31	55.60
0.5	14	40.60	15	34.60	25	54.74	26	51.00
0.6	14	40.60	15	34.60	19	52.84	19	34.50
0.7	11	29.60	12	23.60	16	41.88	16	31.80
0.8	11	29.60	11	27.60	12	31.50	10	17.30
0.9	6	13.80	5	11.80	5	7.70	5	5.60
1.0	0	0.00	0	0.00	0	0.00	0	0.00

Table 3: The entropy-threshold rule on the 50 test cases, at the eleven deciles of each arm’s own $H(b)$ distribution. *ask* is the number of those 50 cases the rule asks about; *+cost* is the realised cost above the shipped policy’s total on the same arm, in cost points, where that total is 86, 86, 70 and 75 across the four columns. Excesses are not comparable across columns for that reason. q is a quantile, so the same q is a different number of bits on each arm; repeated rows within a column are repeated thresholds, not a coincidence.

the eleven-point decile grid of Section 5, on four sets of beliefs: the beliefs this paper reports, those beliefs re-derived through the rebuilt elicitation path, the re-elicited raw beliefs, and the calibrated beliefs of Section 6.9. Every number in this subsection is on those 50 cases.

Each arm’s excess is measured against that arm’s own realised total for the shipped policy, not against one number for all four. The shipped policy scores 86 on the first two arms, 70 on the raw beliefs and 75 on the calibrated ones, so charging the last two an excess against 86 would compare them to a total they never had. That makes the four columns of Table 3 incomparable in magnitude: the smallest entry in the table, +5.60, is the calibrated arm’s excess over its own 75, and it is not a better outcome than the reported arm’s +13.80 over 86. The arms are read down their own columns, at equal quantiles and never at equal bits, because $H(b)$ moves when the belief moves.

No firing threshold is free on any arm. The cheapest is the top decile in every column, at +13.80 on the reported beliefs, +11.80 rebaselined, +7.70 raw and +5.60 calibrated, each against its own reference and each asking about five or six of the 50 cases. The dearest is +128.02, on the raw beliefs at $\tau = 0$, where the rule asks about all 50. The single row at zero excess is $q = 1.0$, and it is zero on all four arms for one reason: the top of the grid is taken over all 100 cases while the scoring is on 50, so τ there sits above every test case, the rule fires on nothing, and what the row prices is the shipped policy under another name. The claim is therefore about firing thresholds specifically. Ten of the eleven rows fire on every arm, and all forty of those cost more than not asking.

The $\tau = 0$ row is the always-ask anchor, and it is priced differently from the always-ask row of Table 2. Asking about all 50 test cases and then taking the action the answer selects costs 199.62, a mean of 3.9924. The committed always-ask total on those same 50 cases is 142, a mean of 2.84 with 21

missed escalations, and it is terminal: it charges the ask and stops. Ask-then-act charges the ask and then the follow-up, so it has to be dearer, and equality on every case would mean the follow-up is free everywhere, which would be a defect and not a result.

One row breaks the pattern that asking less costs less. On the rebaselined arm the excess rises from +23.60 at $q = 0.7$ to +27.60 at $q = 0.8$ while the firing count falls from 12 to 11, and it is the only inversion in the forty-four rows of Table 3. The case that left the firing set is a02-deep-017: the shipped policy answers it and realises 10, where asking and then acting realises 6.0, so dropping it from the firing set moved 4.0 the wrong way. That case’s expected excess is +0.40, still positive. What Theorem 1 bounds is expected cost, and a single realised draw can land on either side of an expectation; here one does. The inversion is a property of the realised column, not a counterexample to the bound, and it is the reason the bound is stated in expectation rather than per case.

Numeric rules and the scale they act on. The decile grid is one of five places in this work where a numeric rule had to be re-expressed relative to the quantity it governs, and the recurrence is the transferable part. A collapse diagnostic written as “fewer than N distinct values” is a raw count, and pooling two halves of the data doubled the count it was compared against while nothing about the distribution changed. A sensitivity sweep written as ± 0.05 and ± 0.10 on a table entry is an absolute probability, and an absolute 0.10 is 14% of a 0.70 entry and all of a 0.10 one. A threshold on $H(b)$ in absolute bits would have put most of its grid where almost no cases are, which is why τ is deciles here. A tie-break resolved by the last bit of a float sum is a rule selecting on a quantity whose tie structure it never consulted: on the 50 test cases the oracle’s $\arg \max_q \text{VoI}$ and the information-gain rule agree on 29, but the oracle needed its tie-break on 13 of them, so the agreement count means nothing without the tie count beside it and is not reported without it. And an annotation offset fixed in points, against a marker radius that grows with the count being printed, gives the worst placement to the bin whose count most needs reading. The five share one shape: a comparison or a placement expressed in units the quantity’s own scale was never consulted for, which measures the grid rather than the phenomenon. Stated as a rule, a numeric threshold, spacing or offset is expressed relative to the scale of what it acts on, or it is reporting the instrument.

6.7 The comparison that is a tie, not a win

Against always-notify, which escalates every message, the cost result is close to a tie: 1.72 against 1.74 over all 100 cases, a total of 172 against 174, and one cost point in eighty-seven on each half of the split. This is not a mistake in the design, it is a property of the cost matrix. A needless notify costs 3 and a false assertion costs 10, so escalating everything is cheap and is barely punished; blanket escalation almost never pays the expensive error. The honest claim here is therefore not that the cost-aware policy is cheaper than always-notify, because on cost it essentially is not. What it buys is human load. Over all 100 cases it reaches the same cost while escalating 43 times rather than 100, at an escalation precision

of 0.605 against 0.420; on the test half alone it escalates 20 of those 50 cases against always-notify’s 50, at 0.65 against 0.42. The same outcome without dragging a person into more than half again as many conversations is the operational win, and it is a different claim from a cost win, so both are reported.

6.8 The development and test split

Both halves reach the same total cost for the cost-aware policy (86 each) and for the uniform baseline (129 each). This looks like clean generalization and is not. The split was stratified by archetype and sub-variant and balanced on the needs-human count, so the two halves are matched by construction and the totals coincide for that reason. The policies do make different decisions on the two halves (the escalation precision differs across them), so the halves are not identical case-for-case; the aggregate totals simply agree because the halves were built to mirror each other. On synthetic data of this kind, matched by design, this figure cannot stand as out-of-sample evidence, and it is not read as such.

6.9 Calibration

Every one of the 16 missed escalations in the 100-case run of Table 2 came from the same place: the belief under-estimated the needs-human probability, not the readiness. On all 16, the needs-human estimate was 0.30 or below against a true label of needing a human, while the readiness argmax on those same cases was spread across cold, hot, and warm and was often correct and confident. The belief reads under-confident on the needs-human marginal, with an expected calibration error of 0.142 (bootstrap 95% CI [0.100, 0.249]; 0.168 on the development half, 0.184 on test), and reliability bins (Figure 1) that read low against observed frequency near the escalation threshold (for example, a predicted 0.10 against an observed 0.40, and a predicted 0.30 against an observed 0.588).

Two qualifications belong with that number. First, the bin that actually *contains* 3/13 is not the under-confident one: at 0.2–0.3, the largest bin at $n = 35$, predicted 0.200 against observed 0.171, the belief is very slightly *over*-confident, and its Wilson interval [0.081, 0.327] covers the predicted value. The under-confidence is in the neighbouring bins. So “under-confident exactly where the threshold sits” is not the right description; the mechanism is that a large mass of cases sits pinned at $b_h = 0.20$, just below the line, and 35% of the case set turns on whether that single elicited value is right. Second, with 4 to 35 cases per bin most individual bins are consistent with perfect calibration, so the ECE point estimate carries most of the evidential weight here and its interval is wide.

Because the failures trace to the needs-human marginal rather than to the readiness marginal, recalibrating that marginal is the natural first fix. Two fits are reported and they are not interchangeable: Section 7 fits on all 100 cases and is an in-sample ceiling, while the fit below is fitted on 50 development cases and scored on the 50 test cases. This is weaker than the claim an earlier draft made. That the misses localise to one marginal is consistent with the independent two-part belief, but it is not evidence *for* independence: the experiment records only the two marginals, never the joint, so it cannot

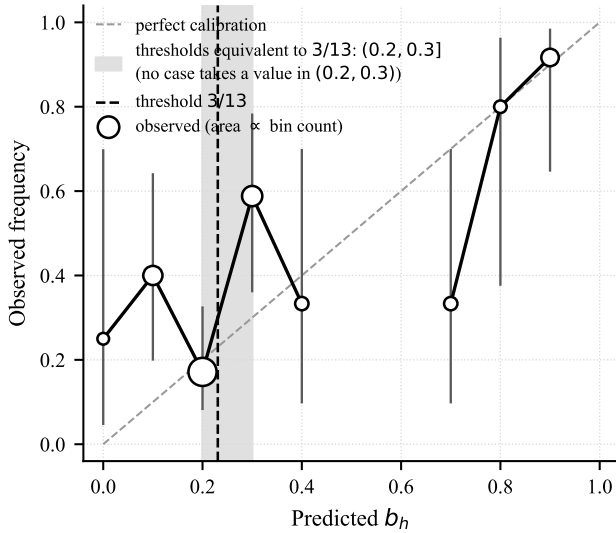


Figure 1: Reliability of the `needs_human` marginal over all 100 cases, ECE 0.142 (bootstrap 95% CI [0.100, 0.249], median 0.171). Points above the diagonal are under-confident: the belief reports a low probability where the observed frequency is high. Marker area is proportional to bin count and bars are Wilson 95% intervals, because the bins hold between 4 and 35 cases and most are individually consistent with perfect calibration. The vertical line is $b_h^* = 3/13 \approx 0.231$; the shaded band is the interval of thresholds that decide identically, which is the half-open $(0.2, 0.3]$. The two brackets are not interchangeable: the elicited marginal takes only one-decimal values, so no case falls in the *open* interval $(0.2, 0.3)$, while 17 of the 100 cases sit at exactly 0.3 and a threshold placed there still puts all 17 on the escalate side. The bin that *contains* the threshold, 0.2–0.3, is the largest ($n = 35$) and is very slightly over-confident (predicted 0.200, observed 0.171, CI [0.081, 0.327]). Its neighbours at 0.1–0.2 and 0.3–0.4 are under-confident by 0.300 and 0.288. The 0.7–0.8 bin reads over-confident by 0.367 on 6 cases, but its interval [0.097, 0.700] covers the predicted value, so the belief is not uniformly under-confident and that bin is not evidence of the reverse. Bins 0.5–0.7 are empty, hence the break in the curve.

detect the two components interfering even if they did. Testing independence would require eliciting $P(s, h)$ directly and comparing it against $P(s)P(h)$, which this run does not do.

A finer belief, fitted on one half and scored on the other. The diagnosis above is a claim about the elicitation, so it can be tested by changing the elicitation. Re-eliciting b_h as an expectation over the model’s digit logprobs, rather than as a probability the model writes out in words, replaces the eight one-decimal values it produced over the 100 cases with 43 distinct scores on the 50 test cases. The elicitor itself was chosen on the development half by cross-entropy: reading the digit distribution reaches 0.9934 bits there against 1.8004 bits for asking the model Yes or No and taking the probability of Yes, a margin of 0.8071 bits. Three candidate maps were then fitted on those same 50 development cases — identity, Platt, isotonic — under a rule fixed before the fit, that an order-preserving map is kept unless a merging one beats it by more than 0.02 bits. Isotonic reached 0.8356 bits against Platt’s 0.9103 and identity’s 0.9934, a margin of 0.0747, so

the merging map was taken and the shipped map does not preserve the ranking of the test scores. Applied unchanged to the 50 test cases, it improves all three proper scoring rules named in advance: cross-entropy 0.8546 to 0.8136 bits, against a label entropy of 0.9815 bits at a base rate of 0.42; ECE 0.1526 to 0.0696; Brier 0.2063 to 0.1962, with the 43 distinct scores merged to 33. On those same 50 cases the policy driven by the calibrated score stops using *answer* at all: its action census goes from $\{\text{answer } 12, \text{escalate-notify } 21, \text{hold } 17\}$ to $\{\text{escalate-notify } 41, \text{hold } 9\}$, so on this arm the two never-used actions of Section 6.3 are three. The trade-off is sharpest in the escalation rates on those 50 cases: precision falls from 0.667 to 0.463 while recall rises from 0.667 to 0.905, so the calibrated belief catches nearly every case that needs a human and pays for it by escalating most of the ones that do not. These are not two findings. The second is caused by the first, and the cause is the map’s range rather than the scores it happened to produce.

Why *answer* disappears. Isotonic regression by PAVA sets each pooled block’s level to that block’s own positive rate. The lowest block of the committed map pools 23 of the 50 development cases, 6 of which are positive, so its level is the probability $6/23 = 0.260870$ — the reachable-score floor of the map, a probability and not a count — and no input can make the map emit a score below it. The interval $[0, 6/23)$ is unreachable rather than merely unpopulated. The boundary the score has to fall below for *answer* to be the cheapest action is $b_h^* = 3/13 \approx 0.230769$, and it lies inside that unreachable interval:

$$\frac{1}{5} < \frac{3}{13} < \frac{6}{23}, \quad (11)$$

an ordering that spans less than 0.061 end to end, which is why all three are carried as exact rationals rather than to two decimal places. Of the 50 test cases, 24 sit below $3/13$ on the raw score and none does after the map. The calibrated arm is therefore a two-action policy by construction rather than by preference, and the same ordering settles the constrained-menu region of Section 6.3, since $6/23$ is above the $1/5$ bound as well. This is a statement about the map’s range and not about how many cases land on it: of the 100 scored cases exactly one sits at the reachable-score floor, and it is a development case, because interpolation lifts the rest of the lowest block strictly above it. The transferable form is that a PAVA-fitted map’s reachable range is bounded below by the positive rate of its lowest pooled block, and that cross-entropy, ECE and Brier never check whether a policy’s fixed thresholds sit inside that range.

What the held-out fit does and does not settle. Table 4 carries the cost and miss numbers for four arms on the test half, and they are secondary: the claim is the three scores above. The fit itself is a fit on one half of a synthetic set scored on the other, and Section 6.8 has already said that the two halves are matched by construction, so it is not evidence that the belief would transfer to a deployment. What it is evidence for is narrower: the improvement in the three scores does not come from fitting the cases it is scored on. The two operations on the population also have to be kept apart. Section 7 fits the marginal on all 100 cases in-sample and halves the misses

Arm, on the 50 test cases	Total	Mean	Miss
re-elicited, uncalibrated	70	1.40	7
re-elicited, calibrated	75	1.50	2
written belief	86	1.72	8
always-notify	87	1.74	0

Table 4: Secondary numbers for the held-out fit, on the 50 test cases only. Total is the realised cost over those 50, Mean is per case, and Miss is missed escalations out of 50. These are reported next to the calibration metrics and are not the claim; the claim is the three proper scoring rules. The test half carries only 8 of the 16 missed escalations of the full 100-case run (Table 2), so a change of one or two misses on this half is not evidence. The written-belief row re-elicits the one-decimal probability of that run and reaches the same total, mean and miss count on this half as its cached beliefs do — aggregate agreement only, not a case-for-case replay. It also shows why the denominators matter: the mean is unchanged from the 100-case run at 1.72 while the miss count is half, so the mean may be compared across the two populations and the count may not. The two re-elicited rows are the calibration contrast, 7 misses of 50 down to 2 of 50 at a mean rising from 1.40 to 1.50 — breakable, and not free. Always-notify escalates all 50 by construction and the calibrated arm escalates 41 of them, so much of what calibration buys here is a move toward blanket escalation.

from 16 to 8; restricting the same policy and the same beliefs from those 100 cases to these 50 also produces 16 to 8, and the two have nothing in common beyond the pair of numbers. Every miss count in this subsection carries the case set it was counted on for that reason.

7 Failure analysis

The cost-aware policy makes 16 missed escalations and no other class of error worth naming: no hard-constraint violations, and its needless actions are the cheap ones the design is willing to pay. So the failure analysis is almost entirely about the misses. Five are examined in detail below, chosen to cover the distinct ways a miss happens.

7.1 The shared cause

All 16 misses have the single cause described in Section 6: an under-estimated needs-human probability. None of them come from the two hidden components being confused with each other. This matters for how the misses should be counted. Treating every miss as a false assertion at cost 10 overstates the damage, because a *hold* is a partial hedge: it is the wrong action, but it does not deliver a false answer, so the matrix charges it 3 or 4 rather than 10. Across the 16 misses the realised costs are nine at 10, four at 3, and three at 4, for 114 in total, not the 160 that costing every miss at 10 would give. One caveat belongs with this diagnosis. That every miss traces to an under-estimated needs-human probability is a statement about this labelling scheme, not a proof of the underlying causal mechanism: the experiment does not independently exclude cost-model misspecification, the coarse three-state readiness space, or the one-decimal quantization of the belief as contributing causes, and the recalibration in Section 6.9 is an in-sample ceiling rather than a held-out test.

Case	Cost	Mode
a11-repeated-097	10	tie-break, not the cost model; since corrected
a04-booking-042	10	readiness right, needs-human wrong
a02-deep-015	10	one quantization step below the line
a03-followup-024	4	hold looked cheap, readiness also off
a10-persistent-091	3	cheapest miss the matrix allows

Table 5: Five missed escalations, one per failure mode. All five are from the reported run. The first has since been fixed in the implementation and is retained because the reported run predates the fix; the other four remain open.

7.2 Five cases

Table 5 lists five misses and the mode each one represents.

The three that realise the full cost of 10 fail in different ways. *a04-booking-042* is the cleanest illustration of the shared cause: the readiness estimate was right and confident, and only the needs-human number failed, so nothing about readiness could have saved it. *a02-deep-015* sits one quantization step below the line: its elicited b_h is 0.20 against a threshold of 0.2308. Describing this as missing by 0.031, as an earlier draft did, reads the gap as a small numerical shortfall, which overstates how close the call was. The belief cannot express 0.24; the next value it can emit is 0.30. Recovering this case requires moving it a full decimal step, not nudging it by three hundredths, and the same is true of the 35 cases that share its $b_h = 0.20$. *a11-repeated-097* is the worst of the three, for a reason that is not the cost number — and, unlike the other four cases in Table 5, it is not an open failure: its cause was identified and repaired, and it is retained here because the reported run predates the repair. Its belief is the furthest from the flip point of any miss: it needed a needs-human estimate above 0.50 and the model reported 0.00, so recalibration would not save it the way it would save the near-threshold cases. And it was not decided by the cost model at all. Two actions came out with an expected cost of exactly 0.000, and the tie was broken by the fixed ordering of the action list rather than by any cost comparison. In the true state, holding would have cost 3, so 7 of its 10 realised points were attributable to an implementation detail, the tie-break, rather than to a property of the policy.

That detail has since been repaired, as Section 5 notes. Ties now resolve safest-first, in an order derived from the cost matrix by worst-case cost (*tie_break_order*) rather than hand-written, which gives *escalate-notify* (worst case 3) < *ask* (4) < *escalate-pause* (6) < *hold* (8) < *answer* (10). The old ordering resolved indifference toward *answer*, the action with the largest downside, which is the opposite of what a policy premised on the expense of false assertion should do. Exactly one decision on the reported case set changes: *a11-repeated-097* moves from *answer* to *hold*, and the mean cost falls from 1.720 to 1.650 — the 7 points this section predicted, and no others. Miss count and escalation

Case	Primary	Secondary
a11-repeated-097	wrong action policy	poor calibration
a04-booking-042	poor calibration	—
a02-deep-015	poor calibration	—
a03-followup-024	poor calibration	wrong threshold
a10-persistent-091	poor calibration	wrong threshold

Table 6: Primary and secondary cause for each of the five failures of Table 5. A dash marks a single-cause failure, not a missing entry. Two mechanisms discussed below carry no label at all: the quantization step behind a02-deep-015, for which the scheme has no entry, and the turn-boundary failure of the last subsection, which is not one of these five and cannot occur in this case set.

count are unchanged at 16 and 43, since *hold* and *answer* are both non-escalating actions. This is also the only decision the correction *could* change: a11-repeated-097 is the one case in the 100 where two feasible actions come out at exactly equal expected cost, and a tie-break is consulted nowhere else. The uniform baseline is unaffected, for a structural reason worth stating — with every non-zero cost set to one, all five actions have worst-case cost 1, so the derived order degenerates to the declaration order and the baseline’s 2.58 is identical under both rules. Table 2 reports the pre-fix numbers, because `results/run.json` was generated before the change and is kept as the artifact the reported numbers are checked against; `run_policies.py --legacy-tie-break` reproduces it exactly, action for action on all 100 cases, so those numbers stay verifiable while the default path uses the corrected rule.

The two cheaper misses show why not every miss is a false assertion. a03-followup-024 and a10-persistent-091 were held rather than answered, so they cost 4 and 3 rather than 10. For these the binding comparison is *hold* against *escalate-notify*, not *answer* against *escalate-notify*, and because the *hold* row is not flat across readiness their thresholds are readiness-dependent rather than the clean 3/13 of the *answer* cases. Reporting them at their true cost, rather than rounding every miss up to 10, is what keeps the miss total honest.

7.3 What each failure is tagged as

Table 6 gives each of the five a primary cause and, where the failure is genuinely two-cause, a secondary. The scheme is a fixed ten-name list: missing hidden state, bad prior, bad evidence, misleading evidence, poor calibration, wrong threshold, wrong cost assumption, distribution shift, insufficient information, and wrong action policy. Two rows carry no secondary, and the blank is deliberate: filling it would assert a second mechanism the artifact does not show.

Poor calibration is primary on four of the five, and on a04-booking-042 it is the whole account: readiness was right and confident, so no second cause is available and none is asserted. a11-repeated-097 is the one row where calibration is not primary. Its calibration error is real and is the largest in the set, but it is non-binding, because the decision was not reached by comparing expected costs at all — two actions tied and the declaration order of the action list resolved it, and the repair described above acts on that order rather than

on the belief. What decided the case is what the primary tag names.

Quantization appears in neither column, and it is the mechanism behind a02-deep-015. The ten names have no entry for a representation limit — a belief whose granularity is coarser than the threshold it is compared against — and filing it under one of them would itself be the failure the numeric-rules paragraph of Section 6.6 names: a comparison expressed in units the quantity’s own scale was never consulted for. The gap has that same shape, which is why it is carried in prose here rather than as a label. The recalibration below shows why it is not a small shortfall: the bin this case sits in has an observed frequency that is itself below the line, so refitting the marginal moves the case away from the threshold rather than across it.

The secondary on a03-followup-024 and a10-persistent-091 is *wrong threshold* and not *wrong cost assumption*. The cost matrix is not in question on either case; what moves is the flip point, because for these two the binding comparison is *hold* against *escalate-notify* and the *hold* row is not flat across readiness, so mass placed on the wrong readiness state raises the bar the needs-human estimate has to clear. On a03-followup-024 the readiness estimate is also wrong, and that is what raises the bar; it does not produce the miss on its own, since the estimate falls short of the raised flip point as well as of the clean one.

One of those two is not stably a failure. The re-elicitation recorded in Section 8 moves a10-persistent-091 out of the miss set, so whether this case is a miss at all depends on which elicitation is read. It is the cheapest miss the matrix allows and is retained for that reason, and the instability is stated here because a tag on this row is a tag on one of two readings.

Read down the table, the five do not scatter across ten independent causes. Primary is *poor calibration* on four and *wrong action policy* on one, which restates the shared cause above rather than adding to it. The diagnostic content is in the secondaries and in the prose, where three distinct mechanisms — the tie-break, the quantization step, and the readiness-dependent threshold — are what turn one under-estimated marginal into a realised cost. The primary column says what failed; it does not say why the failure was expensive.

Two of the ten names are absent by argument rather than by accident. *Distribution shift* is not tagged anywhere: the two halves of the split are matched by construction, as Section 6.8 records, and the only cross-population evidence in this work is the per-case re-elicitation drift of Section 8, which does not appear in any aggregate. Tagging distribution shift would name a cause the artifacts exclude. *Insufficient information* is not tagged either, and there the reason is the central result: on this action menu asking is never rational, by Theorem 1, so a question the agent did not ask is correct policy and not a missed opportunity. A failure analysis reaching for “it should have gathered more” would be contradicting the paper’s own theorem.

The caveat that opens this section governs the whole table. That every miss traces to an under-estimated needs-human probability is a statement about this labelling scheme and not a proof of mechanism, and the tags inherit exactly that

status: they describe what the labels attribute, and they do not exclude cost-model misspecification, the coarse readiness space, or the quantization of the belief as contributing causes.

7.4 What recalibration actually recovers

Since the diagnosis above points at the needs-human marginal, it can be tested directly rather than left as a prediction. Replacing each elicited b_h with the observed frequency of its reliability bin — histogram binning, fit on the same 100 cases, so this is an optimistic in-sample ceiling and not a held-out result — and re-running the policy gives mean cost 1.250 against the corrected baseline of 1.650, and 8 misses against 16.

The composition of that change matters more than the totals. Of the 16 original misses, 9 are fixed and 7 survive. The survivors are the cases the diagnosis already flagged as out of reach: 6 sit at $b_h = 0.20$, in the bin whose observed frequency (0.171) is itself below the threshold, so recalibration moves them the wrong way, and 1 is *all-repeated-097* at $b_h = 0.00$. Recalibration also *creates* one new miss: *a06-frustrated-055*, at $b_h = 0.70$, is pulled down into the over-confident 0.7–0.8 bin and switches from *escalate-notify* to *hold*.

The cost is paid in human load. Over the same 100 cases escalations rise from 43 to 60, escalation precision falls from 0.605 to 0.567, and recall rises from 0.619 to 0.810. Recalibration does not make the policy better on every axis; it buys recall with load, and 8 misses remain that no amount of recalibrating this marginal removes. That residual-miss floor, not the halving, is the finding.

7.5 A failure the test set cannot show

One failure did not come from the synthetic run at all. In a live deployment a lead sent a routine request for information and, before the agent had replied, a second short message asking to be contacted directly by a person. The agent acted on the routine message and silently dropped the second, which carried the only needs-human signal in the turn. The rare, high-cost signal was averaged away by the common, low-cost one because the two messages were handled as a single turn.

This is a different kind of failure from the 16 misses. Those are belief-scoring errors on a single message. This one is a turn-boundary error: the important message never received its own belief evaluation. The synthetic test set cannot produce it by construction, because every case carries one message as a single string and the harness scores one message per turn. Surfacing this class of failure would need multi-message-per-turn cases and a per-message evaluation step before the turn is acted on, neither of which this experiment has, so none of the numbers reported here bear on it. It is recorded here, and as a limitation in Section 8, rather than left out because the harness happens not to reach it. It is also the failure that several practitioners, on being shown it, independently proposed the same fix for: scan each message for a small set of critical intents before the turn is scored as a whole, so that a rare handoff signal overrides rather than averages into the turn.

8 Limitations, ethics, and human control

8.1 Limitations

The results rest on synthetic data, and several of the paper's numbers are only measurable because of that. The clearest case is recall on missed escalations. In a live inbox there is no label for the lead who needed a human and did not get one, so escalation recall cannot be measured at all; it is measurable here only because the cases are labeled by construction. The needs-human base rate of 42% was chosen so that recall is estimable. Earlier drafts described it as inflated above a real inbox; that comparison is withdrawn, because no measured base rate for this channel exists to compare against. What can be said without a source is narrower and still sufficient: the rate was set for measurability rather than sampled, so the precision and recall reported here are properties of this case set and should not be expected to transfer to production, in either direction. The readiness distribution is off the design's own prior in the same way: the belief module assumes 85% cold, the case set is 39% cold. The readiness and needs-human labels are the author's judgment of what each message was written to represent, not outcomes observed from real leads, so they carry the author's assumptions about intent.

Two scope limits on the headline result belong here as well as in Section 6. The cost improvement is against the uniform-cost baseline only. Against always-notify the comparison is a near-tie: over the same 100 cases the two differ by two cost points in 174, 1.72 against 1.74, and by one in eighty-seven on each half of the split, so the cost-aware policy is not the cheaper policy and is not claimed to be. No interval is put on that difference, because none was computed; it is reported as a near-tie and nothing stronger. What it buys is human load — over all 100 cases it escalates 43 times rather than 100, at an escalation precision of 0.605 against 0.420 — and that operational reduction, not a cost win, is the claim. The second is the development and test split. Both halves reach the same total cost, but the split was stratified by archetype and sub-variant and balanced on the needs-human count, so the halves are matched by construction and the agreement follows from that; it is not out-of-sample evidence and is not read as any.

The state space is deliberately coarse and this costs the model in two visible ways. Non-leads such as competitors and abusive messages are labelled cold rather than given their own state, which slightly distorts readiness calibration. And the state cannot separate a competitor fishing for pricing from an abusive message: both come out as cold and needing a human, yet the two call for different actions. The state space simply cannot see the difference, which is a real finding about its granularity rather than a bug in the policy.

Several limitations are structural to the method rather than to the data, and the first is one an earlier draft got backwards. The policy is myopic: it minimises immediate expected cost and carries no machinery that prices the better belief a question buys on the next turn. That much is a true statement about the implementation and stays a limitation of it. What does not survive is the inference drawn from it — that the *ask* action is therefore undervalued here, that the honest handling this design omits would be to compute its value of information, and that *ask*'s absence from the results follows from the horizon.

Section 6.3 computes it. On the unconstrained action menu the value of one more question is at most $-2/13$ at every belief (Theorem 1); posteriors are beliefs, so the bound binds at every node of a lookahead tree and the budgeted family collapses, $W_k = V_{\text{act}}$ for every $k \geq 1$. Asking is not underpriced by the one-step rule. It is priced, and it loses at every depth. That is a withdrawal rather than a qualification, because the claim asserted a direction the analysis reverses.

What remains is narrower, and it belongs here rather than in the results. The collapse is a property of this cost matrix, and the policy never checks the property it depends on. A question can be rational on the unconstrained menu only where $c_F/\nu + c_T/\alpha < 1$; this matrix gives $16/15$, and scaling the *ask* row by $15/16$ would put the ceiling exactly at zero. On a matrix that satisfied the condition, lookahead would be worth having and this design would have nothing with which to take it — there, being myopic would cost what the earlier draft said it costs here. So the statement that survives is not that asking is mispriced but that being myopic is safe on this matrix, for a reason the policy does not verify. Two readings are wrong and worth naming: $2/13$ is a floor on the gap and not an estimate of its size, and the theorem bounds one more question inside a deeper policy on this matrix and this action menu, not the value of interaction design in general. Where a question can pay is the constrained menu, at $b_h < 1/5$ on the hot ray, and Section 6.3 shows that region is empty here — structurally so on the calibrated beliefs, whose reachable range starts above $1/5$, and contingently on the published ones.

The condition is the one statement here that does not mention this dataset, and it is worth reading in the direction of a different deployment. It relates four prices and nothing else: the two fees on the *ask* row, the cost of a needless escalation, and the cost of a false answer. Raise the cost of a false answer, which is what a higher-stakes deployment does, and the second term shrinks. Cheapen the question and the first shrinks, which Section 6.3 already notes. Either move can carry the sum below one, and on that side of the line the optimal policy is no longer never-ask but sometimes-ask, with the crossing given by the same inequality that fails here. Section 6.4 sweeps that crossing directly, by scaling the *ask* row rather than by repricing a false answer, so the two are one transition seen from opposite variables.

8.2 One environment change, three variables

The condition above is stated for this deployment; the brief’s question is what it says about a different one. Take the move from an inbound sales inbox to a higher-stakes setting — an enterprise or regulated channel where a wrong answer is not a lost lead but a compliance exposure — and change one variable at a time, reading each change straight off $c_F/\nu + c_T/\alpha < 1$ rather than re-running anything.

A false answer that costs a hundred times more is a rise in α . It shrinks c_T/α toward zero and carries the sum below one, so the never-ask result of Theorem 1 is exactly the result that breaks: on that side of the line the optimal policy is sometimes-ask, and the crossing is the one Section 6.4 already sweeps by scaling the *ask* row, since raising α and cheapening the question move the same sum from opposite

variables. This is the one change of the three that overturns a headline finding rather than shifting a magnitude.

Evidence ten times more expensive to obtain is a rise in c_F and c_T together. It pushes the sum further above one, so asking stays dominated and stays dominated harder; the direction of the result is unchanged and only its margin grows. The two changes pull opposite ways on the same inequality, which is why a single number governs both and neither needs its own experiment.

Historical data that no longer describes the incoming population is the case the paper does measure, under another name. It is the reweighting of Section 8: the case mix is not the mix the belief module assumes, and importance-weighting toward the assumed prior moves the cost-aware policy from 1.720 to 1.456 and the baseline from 2.580 to 2.526, widening the gap rather than closing it. The direction survives; the magnitude does not. What that buys for a shifted deployment is a bound, not a transfer: a result whose sign is stable under reweighting is one an unreliable historical mix cannot flip, but its size is a property of whichever weighting is chosen, and neither is authoritative.

The three together mark the boundary of what a fixed cost matrix can be trusted to carry. Two of the changes leave the ordering of actions intact and move only what the paper already reports — a margin, a reweighted mean. The third crosses the condition and turns a proof into its negation, and it does so through a single term, which is the whole reason the condition rather than any one experiment is the part worth carrying to another deployment.

None of this is measured, and the reason is worth stating rather than leaving to be noticed. The environment is held fixed throughout, and the three one-variable checks in `experiments/robustness.py` all move a variable inside it rather than across environments: one cell of the cost matrix perturbed with the ordering this paper claims is load-bearing held fixed, a single scalar threshold on the needs-human marginal tuned on the development half and reported on the other, and the case mix importance-reweighted to the author’s own stated readiness prior. Each of those separates a property of the policy from a property of the sample. None of them crosses the condition, because crossing it takes a different cost matrix rather than a perturbed one, and the repricing named above is stated as what would put the ceiling at zero rather than run as an arm.

The value-of-information analysis carries a boundary of its own, and it is worth saying exactly where it falls. Computing the continuation term needs an answer model: a distribution $P_b(u)$ over what the lead would reply and a posterior b^u for each reply. Both are simulated from the design’s own model of the lead, the same one that produced the belief being scored, so a check that the computation agrees with its own definitions is a self-consistency check of the implementation. It is not evidence that the answer model matches how real leads reply, and nothing here measures that: a second turn per case would be new data. What that bounds is narrow, and the narrowness is the point. Theorem 1 does not use the answer model at all — it needs only that the continuation term is non-negative, which follows from Table 1 alone. The quantities that do depend on the answer model are the per-

case ceilings and the count of questions that resolve the state, not the result that *ask* is never selected. What is unvalidated is the model of the reply; what does not rest on it is the bound.

Abstention is priced out under this matrix, and that is a bounded result rather than an option left untested. *Escalate-pause* is dominated by *escalate-notify* in every one of the six columns of Table 1, by between 1 and 3, so a rule that hands the conversation over instead of notifying raises realised cost on every case it touches and cannot lower the count of missed escalations at all (Section 6.3). With *ask* unselected as well, the reported results exercise three of the five actions. The scope of that is the matrix, not the design space. These costs are expert-set rather than measured, and Section 4 shows at least one magnitude is load-bearing, so a matrix in which pausing is not dominated — one that prices an unattended conversation, or a handover acted on sooner than a notification — would make abstention available again, and this experiment says nothing about that case. What it settles is that abstention does not help under these costs, not that it never helps. And the hard constraint that removes the answer action on restricted cases is treated here as observable and error-free, because a flag on the case says the message asks for legal or land documents. A production detector for that condition would carry its own false-positive and false-negative rates, which this experiment does not model. Sweeping a synthetic false-negative rate on that flag from 0 to 0.5 changes nothing — cost stays at 1.650 and misses at 16 — but that is not reassurance, it is a restatement of the fact that the constraint never bound: on all 8 restricted cases the policy already escalates on cost grounds, so dropping the flag changes no decision. A detector’s error rates would matter on restricted cases the cost model does *not* already catch, and this case set contains none.

A further structural limitation is the granularity of the belief itself. The elicited b_h takes only 8 distinct values across the 100 cases, every one at a single decimal place, distributed {0.0: 4, 0.1: 15, 0.2: 35, 0.3: 17, 0.4: 6, 0.7: 6, 0.8: 5, 0.9: 12}. The language model is being asked for a probability and is returning something closer to an ordinal grade. Three consequences follow, and all three are visible in the results: the threshold 3/13 cannot be distinguished from any other value in (0.2, 0.3]; 35% of the case set rests on one elicited value, $b_h = 0.20$, sitting just below the line; and recalibration cannot rescue those 35 cases, because it maps a bin to a single number and that number (0.171) is also below the line. A finer-grained elicitation is the prerequisite for most of the improvements this paper proposes, and it is not tested here. Relatedly, the beliefs come from a single unpinned model (gpt-4o-mini) elicited once per case; no dated snapshot is recorded, no repeat elicitation measures the variance of the belief itself, and every number in the paper is conditional on that one draw. A re-elicitation at temperature 0 returns the same belief for 89 of the 100 cases and a different one for 11, with none unparseable. What the record contains does not determine the cause, and the count is reported here rather than explained.

The eleven are reported case by case rather than as a divergence between the two dates’ beliefs, and the reason is what a divergence cannot see rather than how large one would be. A divergence is a function of the distribution of the values and

is invariant to which case each value belongs to; case identity is where this drift’s whole effect sits. On the test half the decisions that changed moved in both directions and their realised costs cancel to zero, so the totals are the one view in which the drift is not visible and the per-case comparison is the one in which it is. Table 4 draws the same line for the same arm — aggregate agreement, not a case-for-case replay — and any summary that discards case identity discards exactly what that line preserves.

One instrument for the same problem was considered and not built. If the concern is that the evidence pattern arriving this period differs from last period’s, the natural alarm is a symmetric, bounded comparison of the two distributions, and Jensen–Shannon divergence is the standard one: it is defined even where the two distributions do not share support, it is symmetric in its arguments so neither period is privileged as the reference, and it is bounded by one bit, which is what makes a fixed alarm threshold on it mean anything. Nothing here implements it, and nothing here measures whether it would fire. It would also not have helped with the drift reported just above, for the reason given there: a divergence discards case identity, and case identity is where that drift’s whole effect sits. It is recorded as a design option for a deployment that monitors an incoming stream, where the population and not the individual case is the object of comparison, and as a gap in this paper, which has no such monitor.

Reproducibility has a boundary inside the artifact and another outside it. Inside: every number reported here is regenerated from the committed beliefs by the committed code, deterministically, and the data artifacts are compared byte for byte — except two that carry floats whose exactness depends on the interpreter, because compensated summation over floats arrived in CPython 3.12. That version is declared as the floor, and those two files are compared to a tolerance of 10^{-9} rather than exactly. The rendered figure is excluded from the byte comparison outright: a PDF records the version of the renderer that drew it and the moment it was drawn, so its bytes move when neither the data nor the code has. Every value the figure plots is checked against the run artifact instead, which verifies the numbers and not the drawing. Outside: the beliefs are not reproducible from scratch, for the reason just given, so the committed cache is an input to the experiment rather than an output of it, and regenerating it would move every number in this paper.

Because the case set’s readiness distribution does not match the prior the design assumes, the reported means are sensitive to reweighting. Importance-weighting each case toward the module’s own prior (weights: hot 0.087, warm 0.342, cold 2.180) moves the cost-aware policy from 1.720 to 1.456 and the uniform baseline from 2.580 to 2.526 — the gap widens from 0.86 to 1.07. It also reverses the comparison against always-notify, which moves from 1.740 to 1.834. The direction of the main result survives reweighting; its magnitude does not. Since the prior itself is expert-set rather than measured, neither weighting is the authoritative one, and the spread between them is the honest statement of what the case set can support.

Finally, one whole class of failure is invisible to the test set by construction. As Section 7 describes, a high-cost signal

batched with a routine message at a turn boundary cannot occur in a set where every case is a single message scored on its own. This is the sharpest version of the general point that synthetic data trades realism for measurability: here the trade hides not a distorted number but an entire failure mode.

8.3 Ethics and human control

The system decides when a person is pulled into a conversation, and the costs are set to make missing a needed human more expensive than a needless handoff. That is deliberate, but it means the agent will sometimes escalate cases that did not need it, spending human attention as the price of not missing the ones that did. The hard constraint on false legal and land claims is the other side of this: some answers are treated as a line the agent may not cross at any belief, rather than a cost to be traded off, because the harm of a confident false claim on a consequential matter is not the kind of thing a cheaper action elsewhere should be able to outweigh. The experiment prices the decision to escalate and nothing after it. A notification that reaches someone who misreads it, or who is already holding more than they can clear, costs the same ν here as one that lands well, and the cost table has no term at all for who is accountable when the answer was wrong. Those are costs a deployment would have to measure, and this design does not. A deeper point raised in the live discussion is worth recording as a direction the current design does not yet meet. The turn-boundary failure is not only a scanning problem; the underlying issue is that a message was droppable at all. A more robust design would make every inbound item durable, with an explicit acknowledge or close step, so that a request for a human keeps returning until something resolves it, regardless of how a later ranking scores its priority. Priority can be fuzzy; existence should not be. The present design scans for critical signals before the turn is acted on, which reduces the risk, but it does not yet guarantee that no message can be silently dropped, and that guarantee is the right target for a production system.

9 Conclusion and new questions

This paper built a small agent that decides, on each inbound message, whether to answer, ask, hold, or escalate to a human in one of two ways, when the lead’s readiness and whether they need a person are both hidden. The decision rule is a textbook one: minimise expected cost under a belief. The contribution is not the rule but an honest account of what it buys. Making the cost matrix asymmetric cuts mean cost by a third against the uniform-cost version of the same rule, over all 100 cases. That third is against that baseline only. Against always-notify, which simply escalates everything, it does not cut cost at all — 1.72 against 1.74 on the same 100 cases — and the paper says so; what it buys there is reaching the same cost while escalating fewer than half as often. Every missed escalation traces to the needs-human marginal rather than to the readiness marginal, which points at recalibration rather than redesign — though, as Section 6 notes, that localisation is consistent with the two-part belief without being evidence for it, since the experiment never observes the joint distribution.

Three questions follow directly from these results. This version closes one of them, answers most of another, and leaves the third open.

The first is calibration, and it is the one it answers most of. Recalibrating the needs-human marginal in-sample halves the misses on those 100 cases, from 16 to 8, and cuts mean cost from 1.650 to 1.250 — but the composition is 9 fixed, 7 surviving, and 1 newly created, and escalations rise from 43 of 100 to 60 to pay for it. The residual-miss floor is real and it has a specific cause: 6 of the 7 survivors sit at $b_h = 0.20$, in a bin whose observed frequency is also below the threshold, so no mapping of that bin to a single value can rescue them. It is therefore set by the granularity of the elicitation, not by the calibration method — and that last claim is the one thing here that can now be reported as tested rather than asserted. What an earlier draft named as the next step, a finer-grained belief and a held-out recalibration fit, is Section 6.9: eliciting b_h from digit logprobs and fitting the map on 50 development cases takes the misses on the other 50 from 7 of 50 to 2 of 50. That residual-miss floor was breakable, and it broke when the elicitation got finer, which is what the claim predicted. It was not free. The mean rises from 1.40 to 1.50 on that half and the calibrated arm escalates 41 of the 50, because a PAVA-fitted map cannot emit a score below the positive rate of its lowest pooled block and the threshold for answering lies underneath that reachable-score floor. So the granularity claim survives and the bound moves: out of the elicitation, into the calibration map.

The second is the ask action, and it is the one that closes. An earlier draft left it open as a question — whether asking is genuinely useful or merely unpriced — and the answer is neither. It is priced, and it loses. Scoring the action by its one-step value of information is what Section 6.3 does, and what comes out is a bound rather than a measurement: on the unconstrained action menu the value of one more question is at most $-2/13$ at every belief in the simplex (Theorem 1), attained at the all-hot vertex, so there is no belief a deployment could reach at which the question is worth its fee. Nor is that an artefact of the one-step horizon. Posteriors are beliefs, so the same bound holds at every node of a lookahead tree, and the budgeted family collapses: $W_k = V_{\text{act}}$ for every $k \geq 1$. Asking does not fail here because the policy looks only one step ahead. It fails at every depth. *Ask* is selected zero times out of 100 and *escalate-pause* zero times as well, and the five-action set is a three-action set in practice — what this version adds is that the two zeros follow from the cost matrix rather than from the case set.

What transfers is not that number but the condition behind it. A question can be rational at some belief on the unconstrained menu only if $c_F/\nu + c_T/\alpha < 1$: the fee for asking when no human is needed, measured against the cost of a needless escalation, plus the fee for asking when one is, measured against the cost of a false answer, together coming to less than the whole. Here they come to $16/15$, so the condition fails, and it fails by 6.25% — scaling the *ask* row by $15/16$ puts the ceiling exactly at zero. A designer with a cheaper question, a single tap or an inference from metadata already collected, is on the other side of that line, and the same inequality says where the line is. Asking is not useless

in general. On this matrix, and on the unconstrained menu, it is priced out; Section 6.3 also carves out the one place it is not, where the hard constraint removes *answer* and the ceiling turns positive over a region of hot beliefs that no case in this set occupies.

The third is the turn boundary. The one failure the synthetic test set cannot produce is the one that came from a live run, where a high-cost signal was lost when batched with a routine message. Building cases with more than one message per turn, and a per-message scan that lets a rare critical signal override the turn, would let this failure be measured rather than only reported, and would test the durability idea that no inbound message should be silently droppable in the first place.

AI-use statement

Code, data, and the logs referenced in this section are available at <https://github.com/kapardhi03/cost-aware-escalation-agent>.

This project was built with AI assistance used as a tool under the author’s direction, not as an author. The author selected the problem, made and locked every design decision, set the cost values and their ordering, wrote all public discussion contributions personally, and verified every reported number against the committed run artifacts.

AI tools were used for: preparing research terms, search queries, and candidate communities and accounts to engage [tools: Claude, ChatGPT, Gemini]; studying the source papers [tool: NotebookLM]; scaffolding, writing, and repairing code, running the test harness, and executing commits [Claude and Claude Code]; and adversarial review passes of the draft [tools used, logged in `review-record.md`]. Per-decision provenance — what the author set versus what a tool proposed — is recorded at the point each decision was made in `build-log.md`.

Every AI output was checked before use. Where a tool was confidently wrong, the error and its correction are recorded rather than propagated; these are logged in the “Important AI errors” section of `research-file.md`. Two are worth naming: a synthesis that recommended an architecture directly contradicting the locked design, and an unverified bound attributed to a mis-cited source, neither of which was carried into this paper.

The author affirms that no conversation in the discussion record is fabricated; that every reported result comes from a real run of the committed code over the committed data; and that no reference is cited that the author did not read.

A Concept reference

A.1 Belief and inference

A hidden state is the part of the world a decision turns on and the agent cannot read off its input. Here it has two components, and they are separate variables rather than levels of one: *readiness*, which is exactly one of hot, warm or cold, and *needs-human*, which is a property of the moment rather than a fourth readiness level. The pair spans six joint states, and every expected cost in this paper is taken against those six. Section 3 gives the argument for keeping them apart: folded

into a single variable, evidence that a lead is hot would mechanically lower the estimated chance a person is needed, and the hot lead who needs a person is the case the design exists for.

A model that emitted the most likely state, or a single confidence score, would be cheaper to build and would not support this decision. Two of the situations the agent must handle are a ready buyer who needs a person and an unready message a bot should not touch; one score cannot separate them, because they differ on one component and agree on the other. The deeper reason is arithmetic. The action is chosen by expected cost under an asymmetric matrix (Table 1), and an expectation needs a distribution over the states rather than a winner among them. A point prediction discards exactly the spread the matrix charges for.

The belief is what the agent holds over the hidden state given what it has seen: a distribution b_r over hot, warm and cold that sums to one, and a separate scalar $b_h \in [0, 1]$ for needs-human. Under the independence assumption of Section 4 the two multiply into a distribution over the six joint states, which is the object expected cost is computed against. The representation carries a stated limit rather than an implied generality: a posterior that couples the two components is not a product of its marginals and cannot be written in this form, so Section 6.4 works with the full six-state distribution wherever that arises.

The belief before an answer arrives is the prior; the belief after a particular answer u is the posterior b^u . In this agent the distinction is load-bearing in exactly one place, because the shipped policy never buys an answer and therefore acts only on priors. Where posteriors do appear, in the lookahead of Section 6.4, the fact that carries the argument is that a posterior is itself a belief and so a point of the same space, which is what lets a bound proved over the whole space bind at every node of a tree nobody enumerates. The word has a second and unrelated sense here: the readiness mix the module assumes, used in the reweighting of Section 8.

A likelihood is the probability of an observation given a hidden state, and it is the term a Bayes update multiplies the prior by. This agent computes none. The belief is produced by a language model reading the message, so the map from evidence to belief is learned and opaque rather than factored into a prior and a likelihood, which is why calibration and not the update rule is the load-bearing assumption (Section 3). The one likelihood-shaped object in the paper is the answer model of Section 6.4, supplying a distribution over replies and a posterior for each; it is simulated from the same model that produced the belief, and Section 8 states what that does and does not support.

Uncertainty here is not a confidence figure attached to an output. It is the spread of the belief over the six states, and it reaches the behaviour only through the comparison of expected costs: two beliefs with different spreads that put the same expected cost on every action produce the same decision, and no report of uncertainty changes that. Hence the two scales the paper keeps apart, bits for the spread and cost points for the consequences, never compared to one another (Section 5). Uncertainty becomes actionable only when it moves which row of Table 1 is cheapest.

A.2 Entropy and information

Entropy measures how spread a distribution is: zero when one outcome is certain, largest when all are equally likely. $H(b)$ in this paper is the entropy of the belief over the six joint states, so it compresses the agent’s whole uncertainty into one number, and it is defined in Section 5 rather than where it is used because a later section places a threshold on it. What it is not is a cost. It cannot say which action is cheapest, because it is blind to which states are expensive to be wrong about, and a belief concentrated on a costly state is low in entropy and high in risk at the same time.

The bit is the unit entropy is measured in when the logarithm is taken base two: one bit is the uncertainty of a fair coin. Fixing the unit fixes the scale, so $H(b)$ over six states has a ceiling, stated once with the definition in Section 5 and not repeated here. The unit also does the discipline work. A threshold in bits is comparable only to another quantity in bits, which is why τ is the only thing $H(b)$ is ever compared against, and why the sweep of Section 6.6 places τ at quantiles of the observed $H(b)$ instead of at evenly spaced absolute values.

Information gain scores a question by how much uncertainty its answer is expected to remove:

$$IG(q | b) = H(b) - \sum_u P_b(u) H(b^u), \quad (12)$$

starting uncertainty minus the uncertainty expected to remain once the answer arrives, averaged over the answers the question could receive. For Bayes posteriors the quantity is non-negative in expectation, though a particular answer can leave the agent less certain than before. This paper does not select questions with it. The ranking that decides anything here is by break-even price, in Section 6.4, and the ceiling of Section 6.3 is the reason: a question can be highly informative and still not worth its fee, so bits are the wrong currency for the choice this agent faces.

The subtracted term above, $\sum_u P_b(u) H(b^u)$, is the conditional entropy $H(b | U)$: the entropy of the belief after the answer, averaged over which answer comes. Naming it that way makes the identity read as a difference of two uncertainties rather than as a formula. It is also the same shape as the continuation term in the value-of-information expression of Section 6.3, with expected leftover uncertainty standing where expected cost of acting later stands. The shapes match and the units do not, and that difference is the whole of why one of the two decides anything.

Mutual information is the information gain of learning one variable about another, and it is zero exactly when the two are independent. Correlation is the weaker instrument: it is zero whenever a linear relationship is absent, which can happen while a strong dependence remains. The distinction has a concrete place here. Readiness and needs-human are *assumed* independent, and Section 4 is explicit that the run never observes the joint and so cannot check the assumption; no mutual information is computed anywhere in this paper. What is observed is the neighbouring fact in Section 6.4: the question aimed at both halves of the state returns a posterior that is not a product of its own marginals, so an answer can couple two components a prior held apart.

A.3 Divergences

Kullback-Leibler divergence measures how far one distribution is from another by asking what it costs to describe samples from the first while believing the second. It is zero only when the two agree and positive otherwise, which is what makes it serviceable as a gap. It is not a distance: it is asymmetric, it obeys no triangle inequality, and it is unbounded, growing without limit as the believed distribution approaches certainty against an observation that disagrees. In this paper it appears as the miscalibration term of the cross-entropy split described in Appendix B, taken per reliability bin between the observed frequency and the confidence stated for that bin.

Because the two arguments are not interchangeable, using KL means choosing which distribution is the reference, and that choice is a modelling decision rather than a convention. Here the observed frequency is the reference and the stated confidence is the argument, so the penalty grows without bound as a confident claim meets a disagreeing observation. The direction is chosen to match the cost matrix: confident and wrong in the needs-human direction is precisely what produces a missed escalation, the error Table 1 prices highest. A symmetric measure would price both directions of error alike, which is the assumption this design is written against.

Jensen-Shannon divergence is the symmetrised, bounded relative of KL: it compares each distribution against their mixture, is unchanged if the arguments swap, stays defined where the two do not share support, and is bounded by one bit. Those are the properties a fixed alarm threshold needs, which is why Section 8 names it as the natural instrument for comparing one period’s evidence against another’s. Nothing here implements it, and the same section states why it would not have helped with the drift actually observed: a divergence is a function of the distribution of the values and is invariant to which case each value belongs to, and case identity is where that drift’s whole effect sits.

A.4 Calibration

A belief is calibrated when the frequencies match the claims: among the cases given a stated probability, about that fraction turn out positive. It is the load-bearing assumption of this design, because the policy compares expected costs computed from b_h , and a systematically low b_h makes the automated actions look cheaper than they are. The paper measures it as expected calibration error on the needs-human marginal, that being the marginal the threshold acts on, with cross-entropy and Brier score reported alongside for the held-out fit of Section 6.9. What happens there is the reason the concept has to be stated carefully rather than assumed: a map fitted on one half improved all three of those measures on the other half while removing an action from the policy’s reach, so a belief can be better calibrated by every rule named in advance and worse as an input to this decision. Appendix B takes up the mechanism.

A.5 Decision and the value of information

The policy scores each feasible action by its expected cost under the current belief, summing that action’s cost in each of the six joint states against the state’s probability, and takes

the cheapest. A threshold is not an extra parameter in such a scheme; it is where two of those expectations cross. Because the *answer* and *escalate-notify* rows of Table 1 are flat across readiness, their comparison reduces to a condition on b_h alone, and Section 4.4 solves it for the crossing b_h^* . The general shape is worth separating from this matrix: the break-even is the cheap error’s cost over the sum of both errors’ costs, so the threshold moves when the prices move and in no other way.

The value of information of a question is what knowing the answer saves, measured in the units of the decision rather than in bits: the cost of acting now, less the fee for asking and the expected cost of acting on whatever posterior the answer produces. Written that way it can be negative, and on this matrix it always is. Section 6.3 proves a negative ceiling at every belief, and because every cost in Table 1 is non-negative the bound survives any depth of lookahead, so the tree collapses to acting now. The form that transfers is the inequality $c_F/\nu + c_T/\alpha < 1$, relating the two fees on the *ask* row to the cost of a needless escalation and to the cost of a false answer; it fails here, and Section 6.4 sweeps the crossing directly. Information is worth buying when it is cheap against the errors it prevents, not because it is informative.

The two properties come apart, and this paper contains the clean case. Section 6.3 reports case-question pairs on which the answer resolves the hidden state outright, leaving the continuation term at zero because whatever is done afterwards is free. Asking still loses on those pairs, and loses by the full ceiling rather than by a residue, because the fee is paid before the answer arrives. A question with maximal information gain is therefore turned down on price. The mirror case is measured too: the control question, whose answer cannot depend on the state, carries a break-even of exactly zero in Section 6.4, worth buying at no price above free. Between them they mark out why the selection rule in this agent compares costs and not bits.

A.6 Shift and stopping

Distribution shift is the case where the population the agent meets stops matching the one its thresholds were set against, and it disables a policy without breaking a line of its code. Two forms appear in this artifact and neither is monitored. Section 8 reports a re-elicitation of the same beliefs from the same model returning a different value on some cases, where the totals cancel and the per-case comparison is the only view in which anything is visible; and it records a population-level alarm, considered and not built, together with the unit its threshold would have to be read in. The detection question is what the object of comparison is. A summary over a stream can see a population move and cannot see a case change identity, and a policy whose thresholds are fixed rationals is exposed to both.

Stopping is a decision like the others here, priced rather than assumed. An agent that can ask, hold or act needs a rule for when further evidence stops being worth its cost, and the value-of-information expression is that rule: gather while the expected saving exceeds the fee, act once it does not. On this matrix the rule returns act-now everywhere, and Section 6.3 shows the answer does not depend on the horizon, so being

myopic costs nothing here for a reason the policy itself never verifies. Section 8 is explicit that this is a property of four prices rather than a general licence: on a matrix satisfying the asking condition the same rule would say gather, and this design would have nothing with which to do it.

B Concepts the results forced

B.1 The reachable-score floor

This one came from a number that looked wrong rather than from the reading list. After the held-out recalibration the calibrated arm escalates 41 of the 50 test cases, and a policy that routes most of its traffic to a human is not obviously an improvement on one that routes all of it. Calibration is usually treated as a fix, so the question was what a fix means when it moves the policy toward always-notify. The answer was not in the cost model. It was in the structure of the fitting algorithm: pool-adjacent-violators sets each pooled block’s level to that block’s own positive rate, which gives the fitted map a lowest value it can emit, and the threshold the belief has to fall below for *answer* to win sits underneath it.

Section 6.9 computes that floor for the committed map, names it, and states the form that transfers, which is the one worth keeping: the bound is the positive rate of the lowest pooled block, so it is a property of the algorithm, while its value here is a property of this fit on this development split. Neither is restated in this appendix. What belongs here is only how it was reached, because the route is part of the concept. No scoring rule in use reports it. Cross-entropy fell and Brier fell while an action was being removed from the policy’s reach, and the two facts are compatible because those rules are computed on the scores and the constraint is on the range the scores live in. Reading the loss would not have surfaced it. Asking why an action count went to zero did.

B.2 The calibration gap is a KL divergence

Calibration and cross-entropy were in the design from the start. Naming what the gap between them actually is came later, and from this project’s own code rather than from the reading. Splitting cross-entropy over a set of reliability bins gives two terms: the label entropy inside a bin, which no predictor can remove, and the gap between the bin’s observed frequency and the confidence stated for it, which a perfect recalibration would drive to zero. The second term is a Kullback-Leibler divergence. `ce_decomposition` in `src/calibrate.py` computes it under that name, as a per-bin Bernoulli KL from the observed rate to the stated confidence, weighted by bin occupancy, and both terms are committed per arm in `results/logprob-elicitation.json`. No number from that decomposition is repeated here.

Naming it changes how the reliability diagram reads. Figure 1 plots stated confidence against observed frequency bin by bin, and the visible gap in the low bins is that divergence. KL is the right instrument for it rather than a symmetric distance, for two reasons that are properties of the decision and not of the plot. First, it is asymmetric, and the asymmetry runs the correct way: taking the observed frequency as the reference makes the penalty grow without bound as the stated

confidence approaches certainty against a disagreeing observation, so confident and wrong is punished hardest, and confident and wrong in the needs-human direction is exactly what produces a missed escalation. The implementation clips only the stated side, which is what keeps the term finite. Second, it is the part of cross-entropy that recalibration can act on, held apart from the part no predictor can, rather than an extra diagnostic laid beside the loss. That separation is not an identity: the two terms recover the measured cross-entropy only if the scores are constant inside a bin, which they are not, so `ce_decomposition` returns a residual next to them and the committed runs carry it. A symmetric distance would price the two directions of error the same, which is the assumption this paper is written against.

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